PIER

MAJOR ADVANCEMENTS IN MACHINE LEARNING

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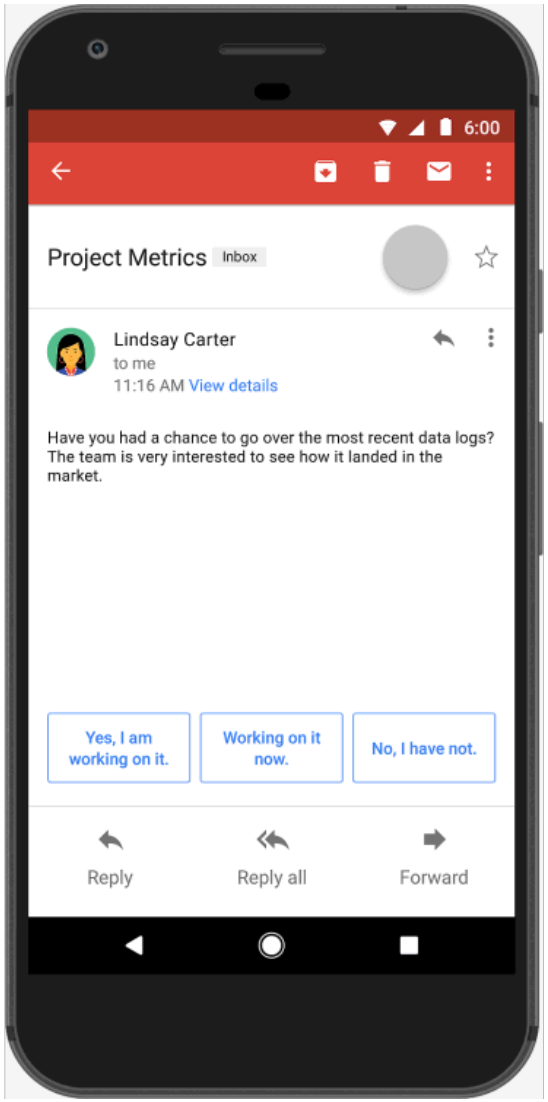
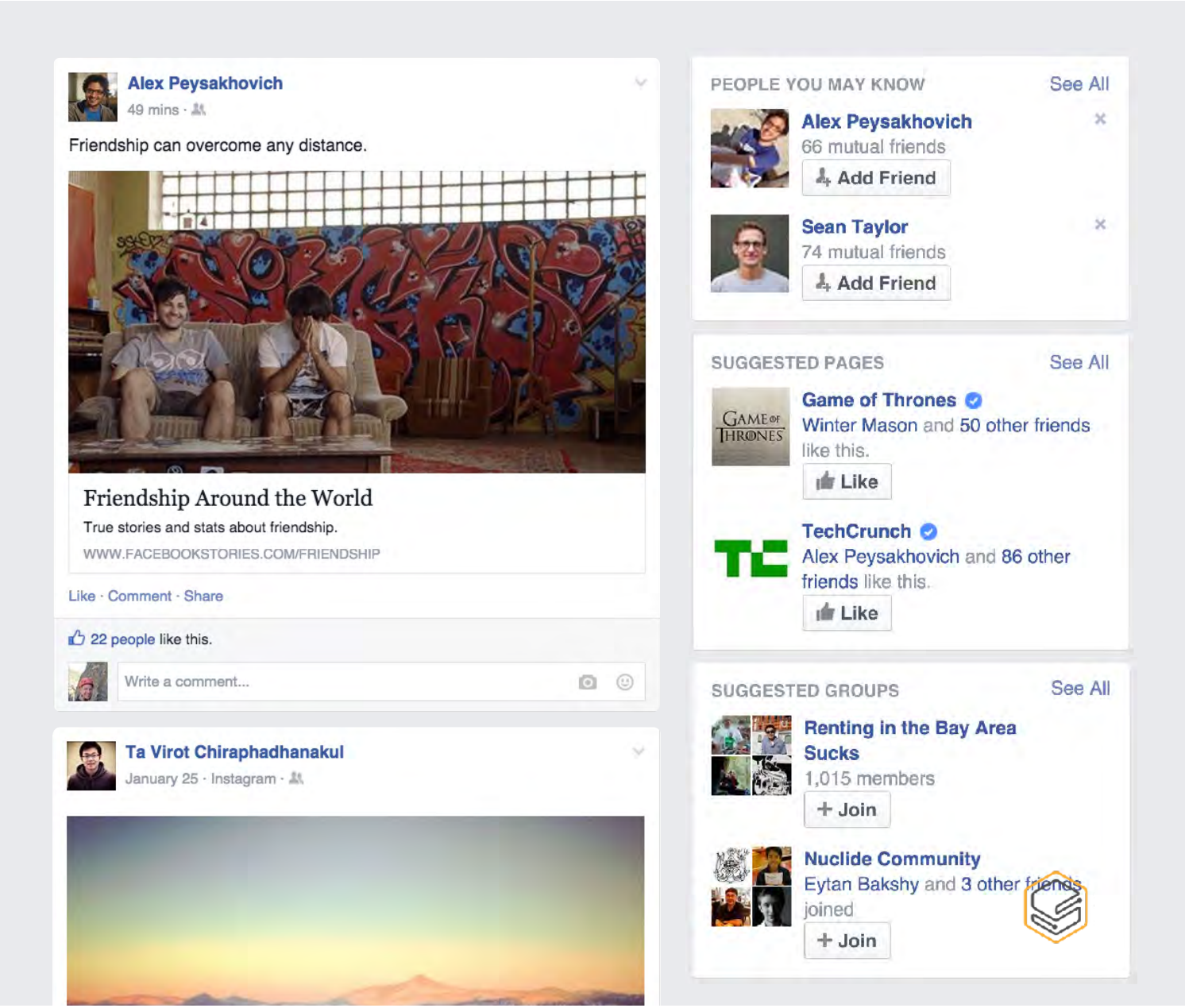
ex-Data Scientist, **Facebook**

Google Developers Expert in Machine Learning
Managing Director @ **Skooldio**
Lecturer @ **Chula Business School**

12:41
Hey Siri what is machine learning
Here's some information:
KNOWLEDGE
Machine learning
Branch of statistics and computer science, which studies algorithms and architectures that learn from observed facts
Machine learning is a subset of artificial intelligence in the field of computer science that often uses statistical techniques to give computers the ability to "learn" with data, without being explicitly programmed.The name machine learning was coined in 1959 by Arthur Samuel.
Follows
Pattern recognition
See More on Wikipedia

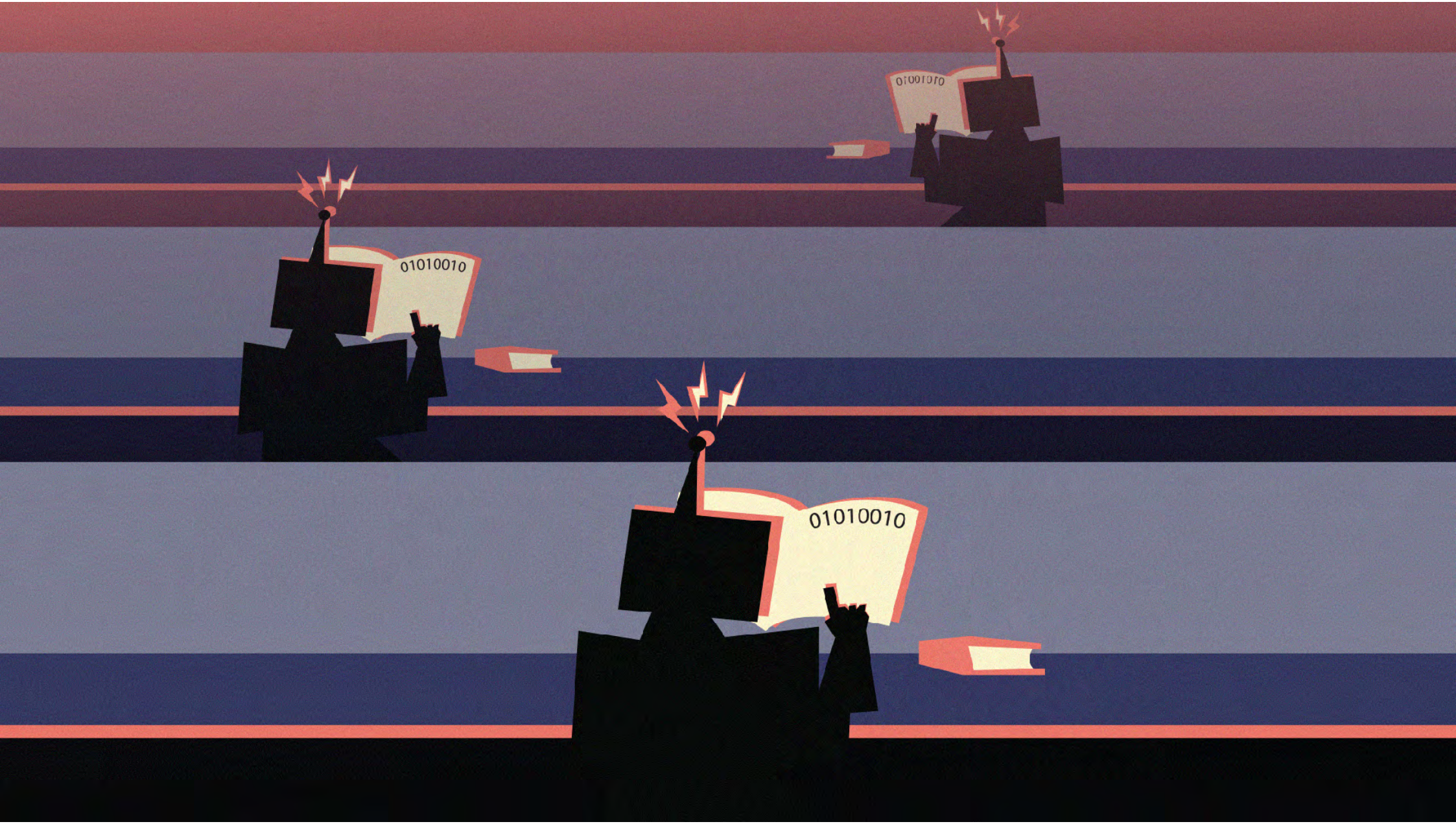

6:00
Project Metrics
Inbox
Lindsay Carter
to me
11:16 AM View details
Have you had a chance to go over the most recent data logs? The team is very interested to see how it landed in the market.
Yes, I am working on it.
Working on it now.
No, I have not.
Reply
Reply all
Forward

facebook


Alex Peysakhovich
49 mins ·
Friendship can overcome any distance.
Friendship Around the World
True stories and stats about friendship.
WWW.FACEBOOKSTORIES.COM/FRIENDSHIP
Like · Comment · Share
22 people like this.
Write a comment...
Ta Virot Chiraphadhanakul
January 25 · Instagram ·
PEOPLE YOU MAY KNOW
See All
Alex Peysakhovich
66 mutual friends
Add Friend
Sean Taylor
74 mutual friends
Add Friend
SUGGESTED PAGES
See All
Game of Thrones
Winter Mason and 50 other friends like this.
Like
TechCrunch
Alex Peysakhovich and 86 other friends like this.
Like
SUGGESTED GROUPS
See All
Renting in the Bay Area Sucks
1,015 members
+ Join
Nuclide Community
Eytan Bakshy and 3 other friends joined
+ Join

> “Artificial Intelligence is the new electricity.”

— Andrew Ng


01010010
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What is Machine Learning?

- A subfield of Artificial Intelligence (AI)
- The field of study that gives computers the ability to learn **without being explicitly programmed** — Arthur Samuel
- Roughly speaking, machine learning is using data (training) to answer questions (predictions)

Supervised Learning

Almost all of AI's recent progress is through supervised learning.

What Machine Learning Can Do

A simple way to think about supervised learning.

INPUT A	RESPONSE B	APPLICATION
Picture	Are there human faces? (0 or 1)	Photo tagging
Loan application	Will they repay the loan? (0 or 1)	Loan approvals
Ad plus user information	Will user click on ad? (0 or 1)	Targeted online ads
Audio clip	Transcript of audio clip	Speech recognition
English sentence	French sentence	Language translation
Sensors from hard disk, plane engine, etc.	Is it about to fail?	Preventive maintenance
Car camera and other sensors	Position of other cars	Self-driving cars

SOURCE ANDREW NG

© HBR.ORG

If a typical person can do a mental task with less than one second of thought, we can probably automate it using AI either now or in the near future.

Source: https://hbr.org/2016/11/what-artificial-intelligence-can-and-cant-do-right-now

Supervised Learning

- Predict the value of y from observed characteristics x based on samples (y_i, x_i)
- For a loss function $L(\hat{y}, y)$, search for a function $\hat{f}$ that has low expected prediction loss $E_{(y,x)}[L(\hat{f}(x), y)]$ on **new** data from the same distribution

Challenges

- Improve prediction accuracy with a more complex, flexible model
- Avoid overfitting; find a model that works well out-of-sample
- Learn from unstructured data (documents, audios, images, videos)

Linear Regression

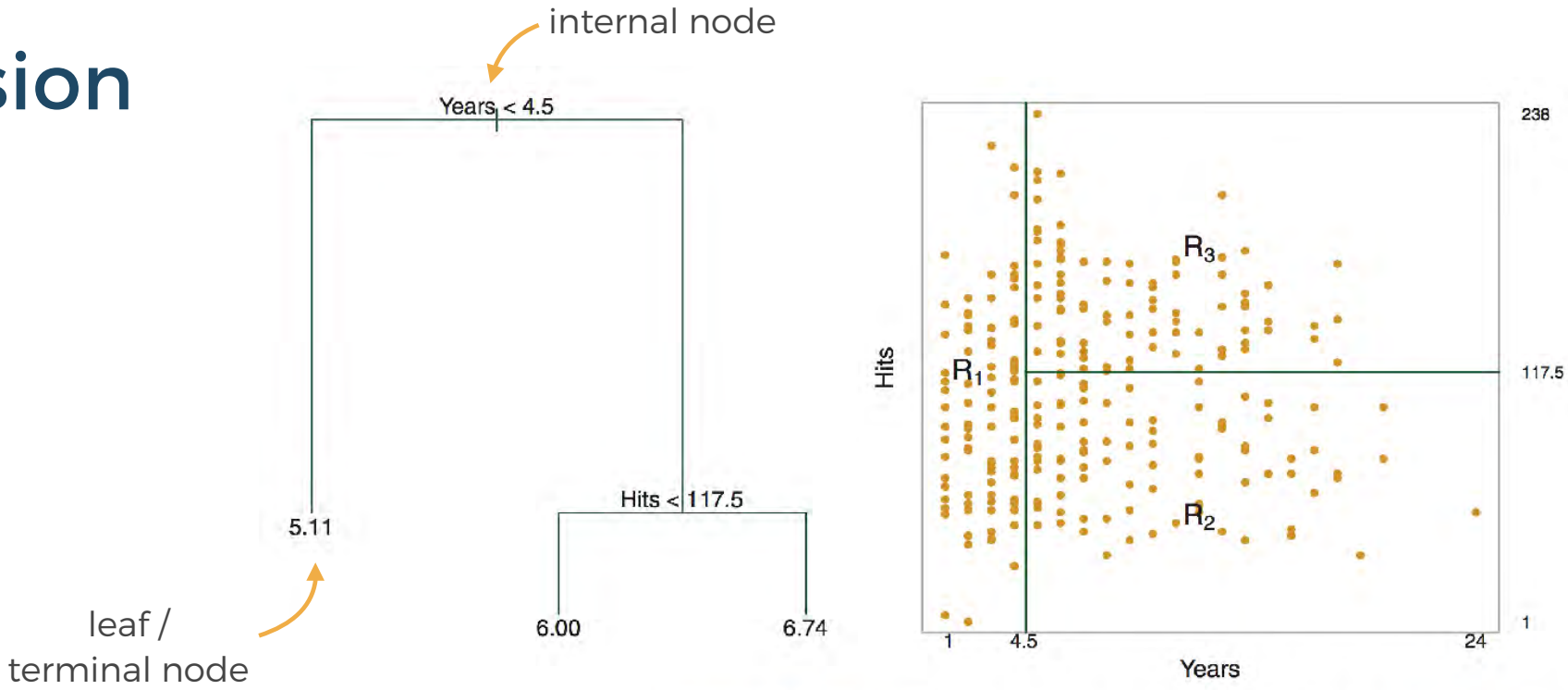
- One of the most useful and widely used method for predicting quantitative responses
- It can be used to understand relationships between the dependent variable (response) and independent variables (predictors, features)

Complex Interactions

Regression Trees

- Divide the feature space into non-overlapping regions
- For every observation that falls in a given region, the prediction is the **same** and given by the **mean** of response values from the training observations in that region

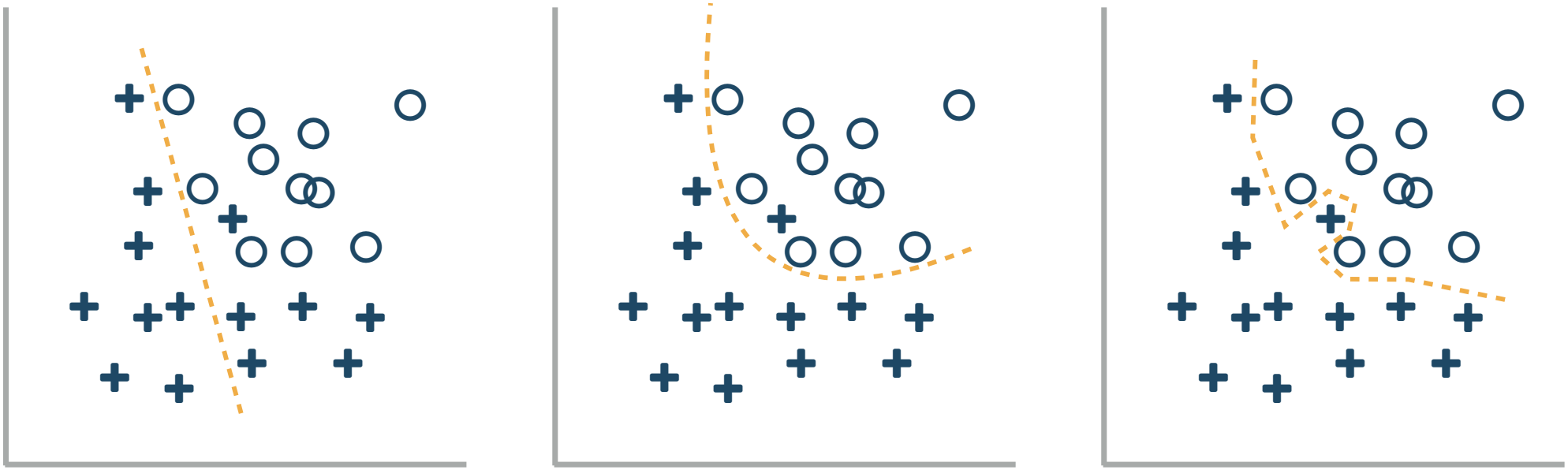
Regression Trees



The figure is taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani

Overfitting

Overfitting



Underfit

Normal

Overfit

Overfitting

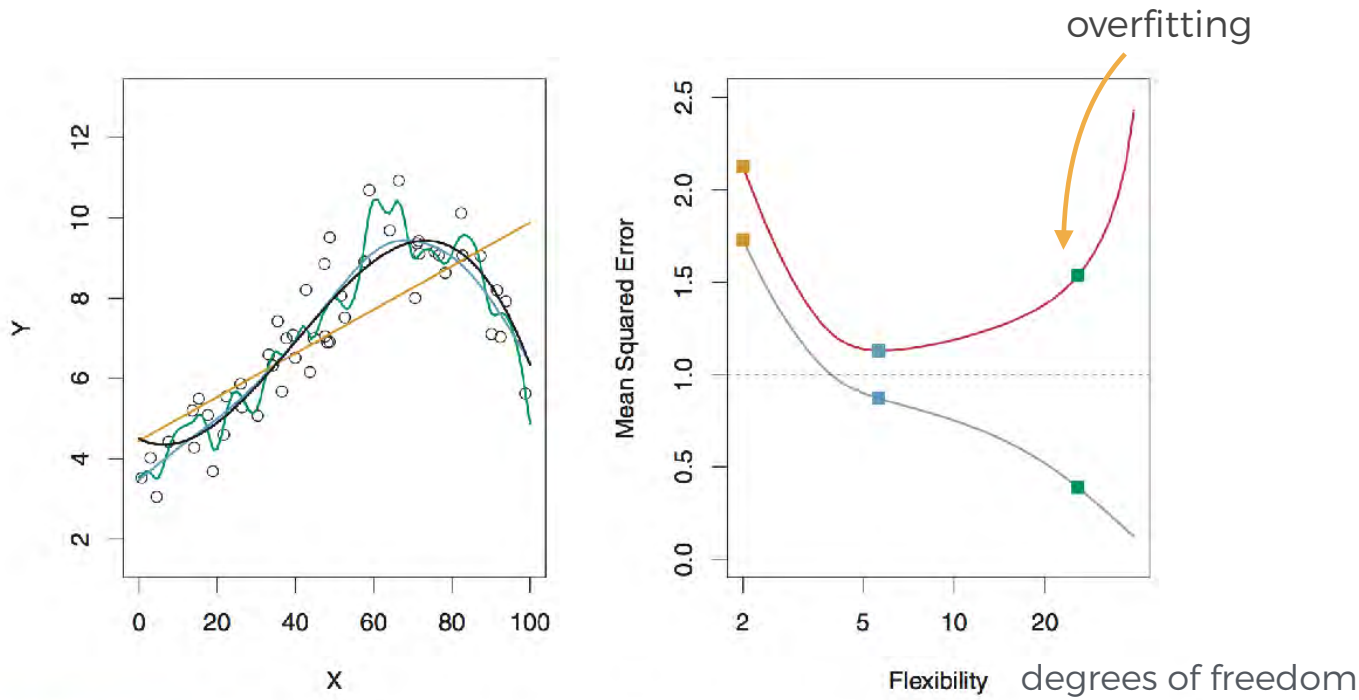


FIGURE 2.9. Left: *Data simulated from f, shown in black. Three estimates of f are shown: the linear regression line (orange curve), and two smoothing spline fits (blue and green curves).* Right: *Training MSE (grey curve), test MSE (red curve), and minimum possible test MSE over all methods (dashed line). Squares represent the training and test MSEs for the three fits shown in the left-hand panel.*

The figure is taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani

Bias-Variance Tradeoff

- **Variance** refers to the amount by which $\hat{f}$ would change if we estimated it using a different training data set
  - Generally, **more flexible** statistical methods have **higher variance**
- **Bias** refers to the error that is introduced by approximating a real-life problem by a simple model
  - Generally, **more flexible** methods result in **less bias**

Bias-Variance Tradeoff

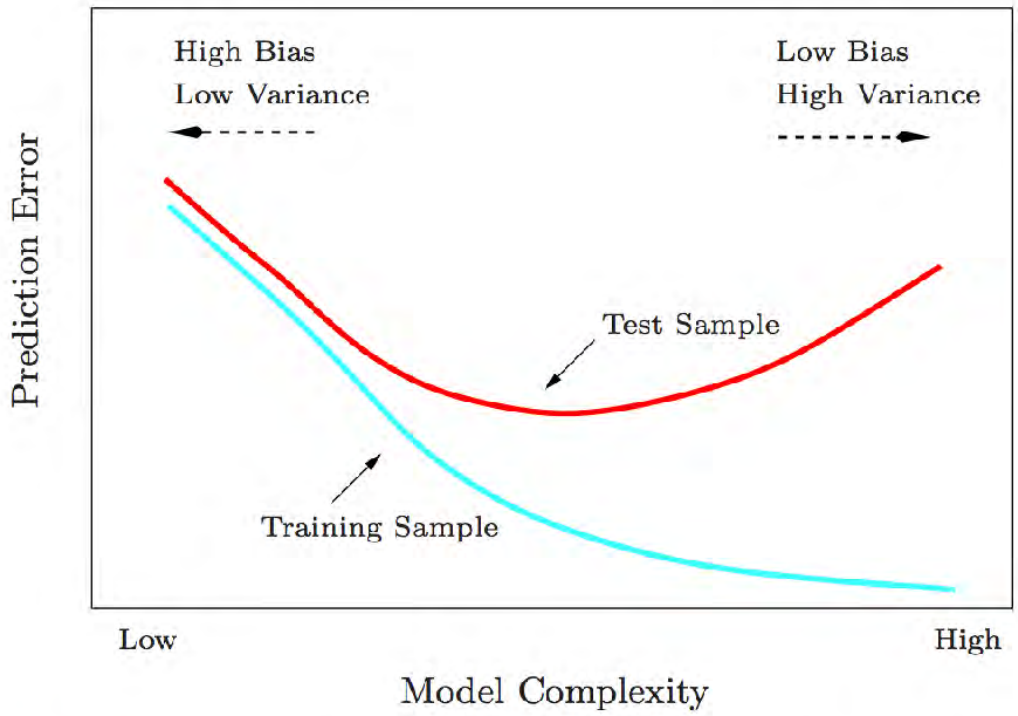


FIGURE 2.11. *Test and training error as a function of model complexity.*

Source: Elements of Statistical Learning (second edition)

Estimating Test Errors

- We are interested in how well our model predicts the response on a new observation
- However, we don't always have a large designated test set
- To overcome this problem, we can estimate the test error rate by **holding out** a subset of the training observations from the fitting process, and then applying the model to those held out observations

K-fold Cross Validation

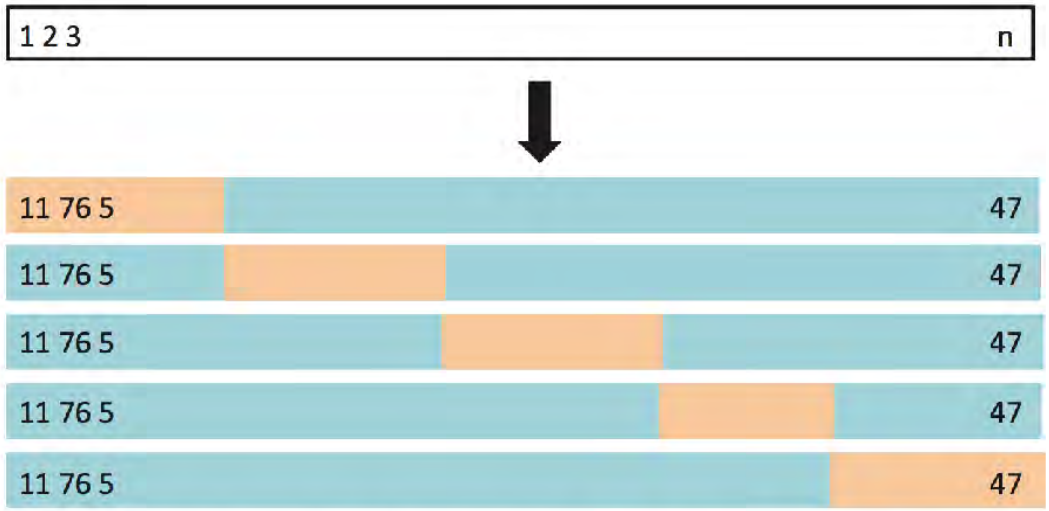


FIGURE 5.5. *A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.*

The figure is taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani

Regularization

Regularization

- aka **Shrinkage methods**
- Shrink the coefficient estimates towards zero to reduce the variance
- Penalize model complexity

The Lasso

- Minimize

tuning parameter

$$\underbrace{\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2}_{\text{training loss}} + \underbrace{\lambda\sum_{j=1}^{p}|\beta_j|}_{\text{regularization / shrinkage penalty}} = \mathrm{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|$$

The Lasso

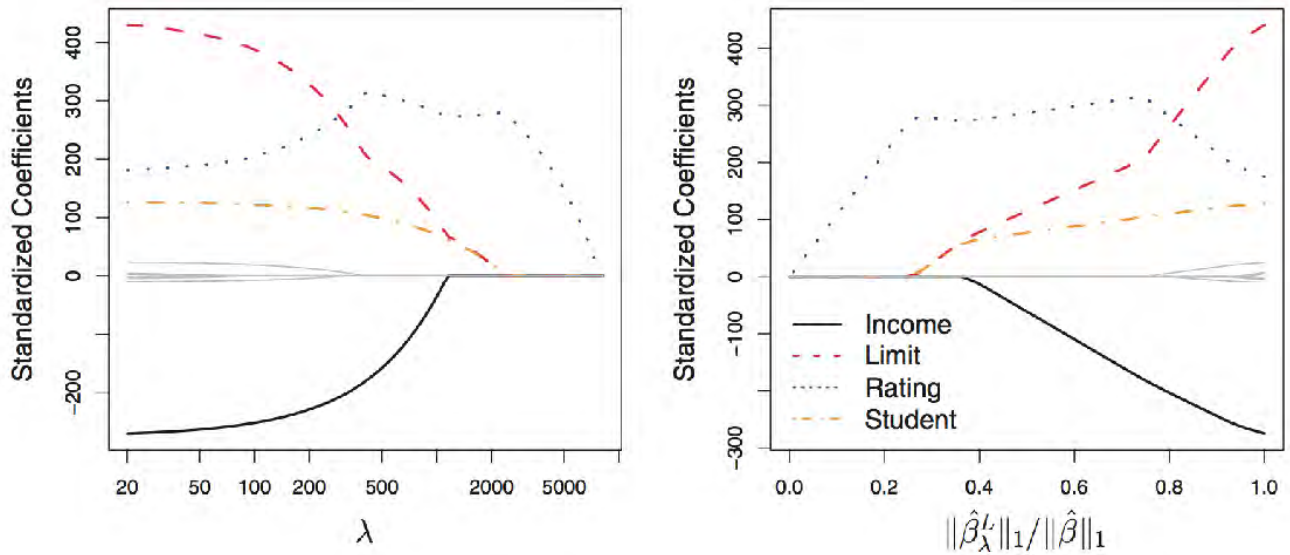


FIGURE 6.6. *The standardized lasso coefficients on the* `Credit` *data set are shown as a function of* λ *and* $\|\hat{\beta}^L_\lambda\|_1/\|\hat{\beta}\|_1$.

The figure is taken from "An Introduction to Statistical Learning, with applications in R" (Springer, 2013) with permission from the authors: G. James, D. Witten, T. Hastie and R. Tibshirani

Regularization

Some Machine Learning Algorithms

Function class $\mathcal{F}$ (and its parametrization)	*Regularizer $R(f)$*
Global/parametric predictors	
Linear $\beta' x$ (and generalizations)	Subset selection $\|\|\beta\|\|_0 = \sum_{j=1}^{k} 1_{\beta_j \neq 0}$
	LASSO $\|\|\beta\|\|_1 = \sum_{j=1}^{k} \|\beta_j\|$
	Ridge $\|\|\beta\|\|_2^2 = \sum_{j=1}^{k} \beta_j^2$
	Elastic net $\alpha \|\|\beta\|\|_1 + (1-\alpha) \|\|\beta\|\|_2^2$
Local/nonparametric predictors	
Decision/regression trees	Depth, number of nodes/leaves, minimal leaf size, information gain at splits
Random forest (linear combination of trees)	Number of trees, number of variables used in each tree, size of bootstrap sample, complexity of trees (see above)

Machine Learning: An Applied Econometric Approach (Mullainathan and Spiess, 2017)

Empirical Tuning

1. Conditional on a level of complexity, determine the best in-sample loss-minimizing function

$$\underset{f \in F}{\text{minimize}} \sum_i L(f(x_i), y_i))$$

$$\text{subject to} \qquad R(f) \leq C$$

2. Estimate the optimal level of complexity through cross validation

Ensembles

Model Averaging

- Decision trees suffer from **high variance**
- We can overcome this by
  1. take many training sets from the population
  2. construct decision trees — one for each training set, and
  3. average the resulting trees

Bagging — Bootstrap Aggregating

- A general approach for reducing the variance of a statistical learning method
- Repeatedly generate **B** different bootstrapped training data sets from the original data set
- Construct **B** trees, one for each bootstrapped training data set
- For regression, **average** the resulting predictions. For classification, take a **majority vote** — the most commonly occurring class among the **B** predictions.

Random Forests

- An improvement over bagged trees by way of a small tweak that **decorrelates** the trees
- When there is a very strong predictor in the data set, most of the bagged trees will use it in the top split
- Consequently, most of the bagged trees look quite similar and are highly correlated
- This limits the reduction in variance over a single tree

Boosting

Can a set of **weak learners** create a single **strong learner**?

- Trees are grown **sequentially** using information from previously grown trees
- No bootstrap sampling. Each tree is fit on a **re-weighted** version of the original data set to **focus on errors.**
- The new decision tree is then added into the fitted function in order to update errors
- In general, **Boosting** > **Random Forests** > **Bagging** > **Single Tree**

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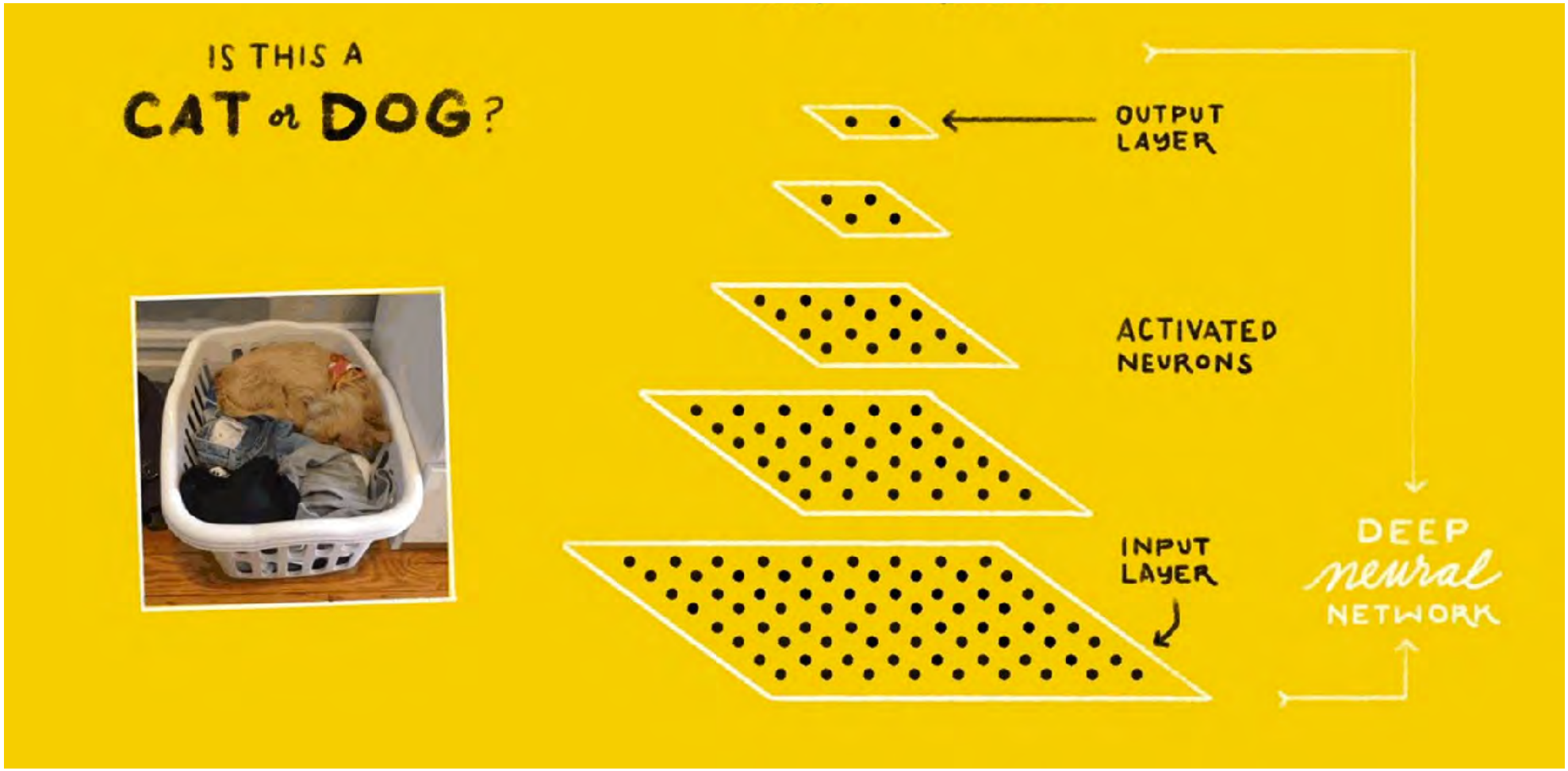
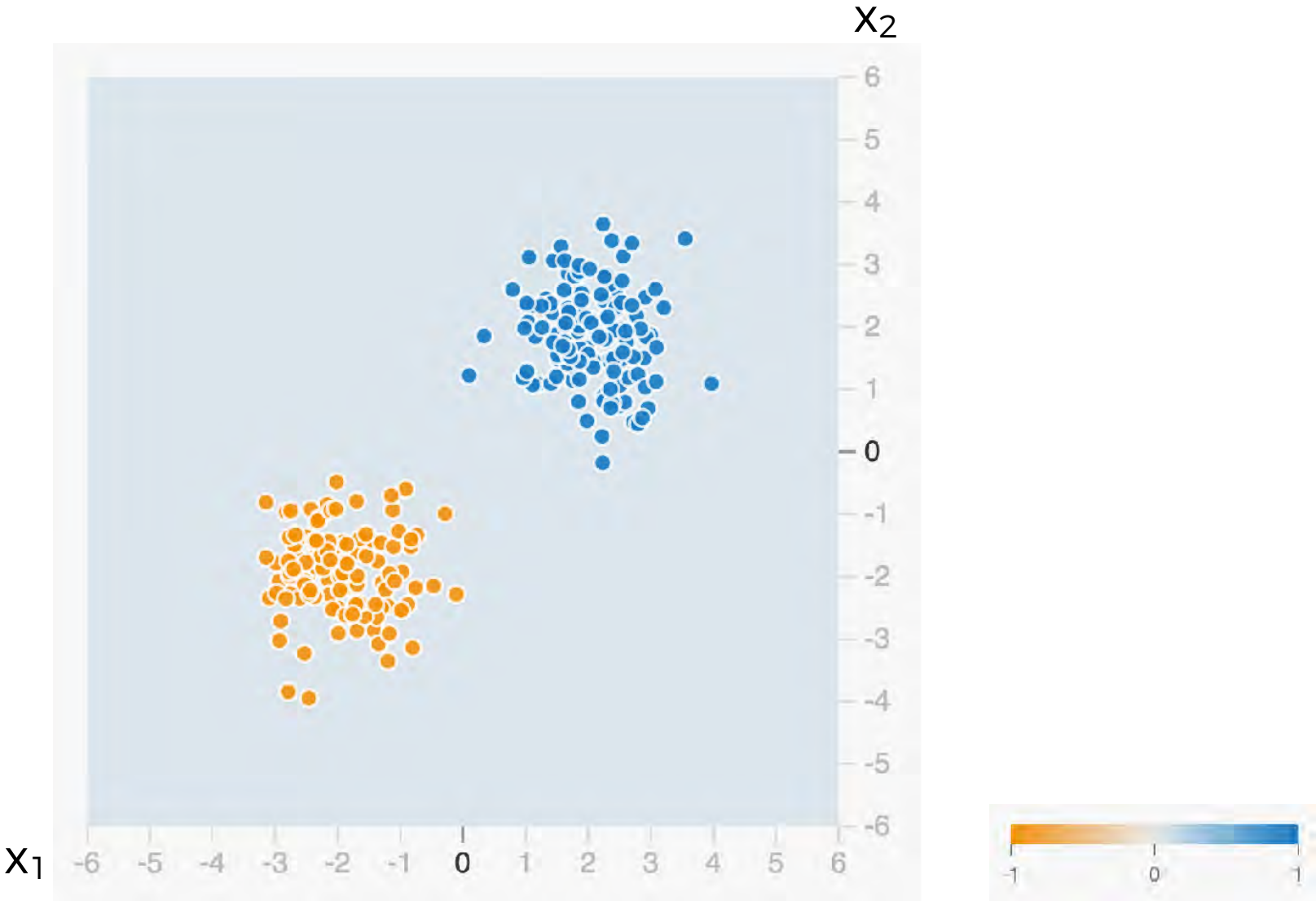
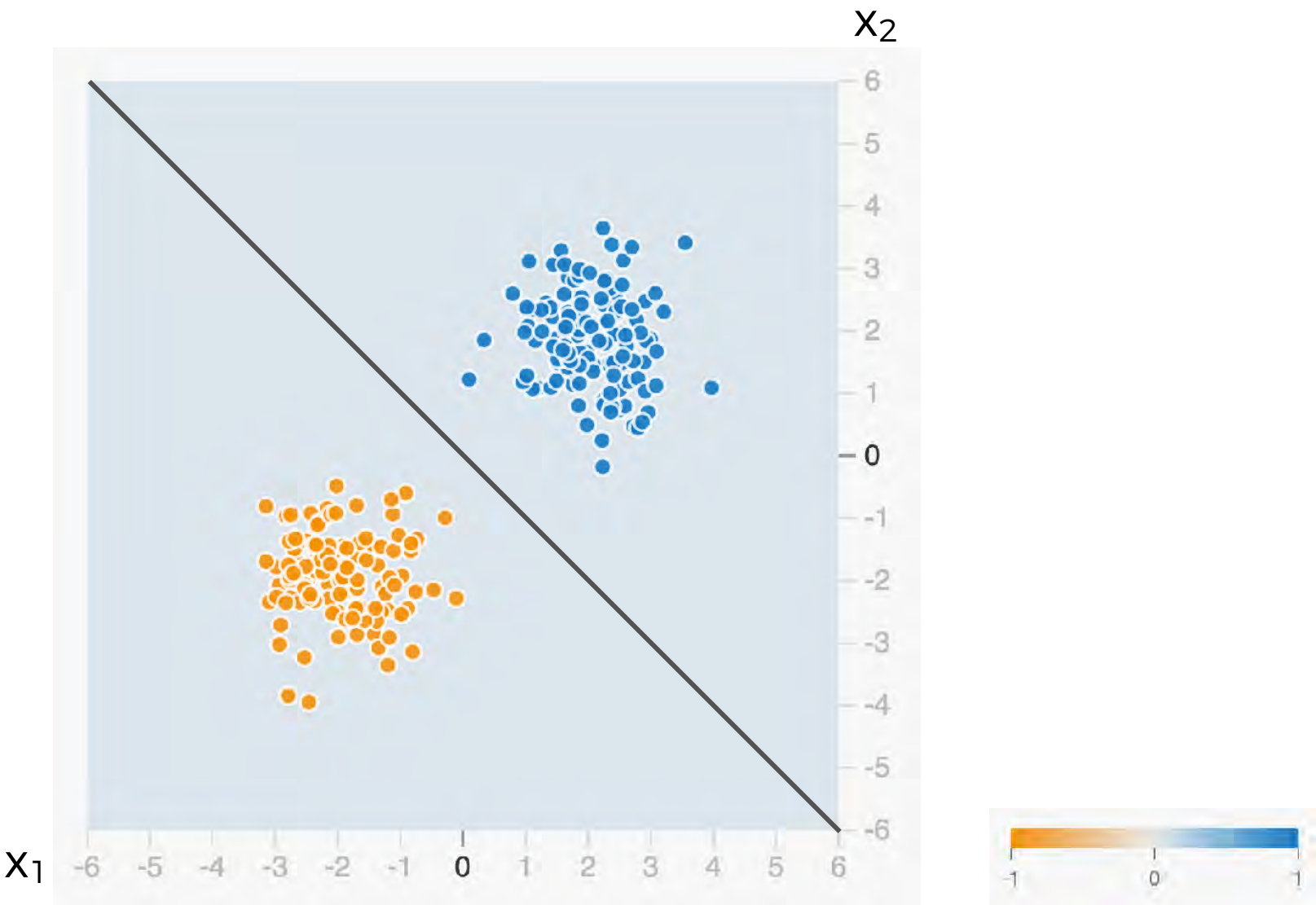
DEEP LEARNING

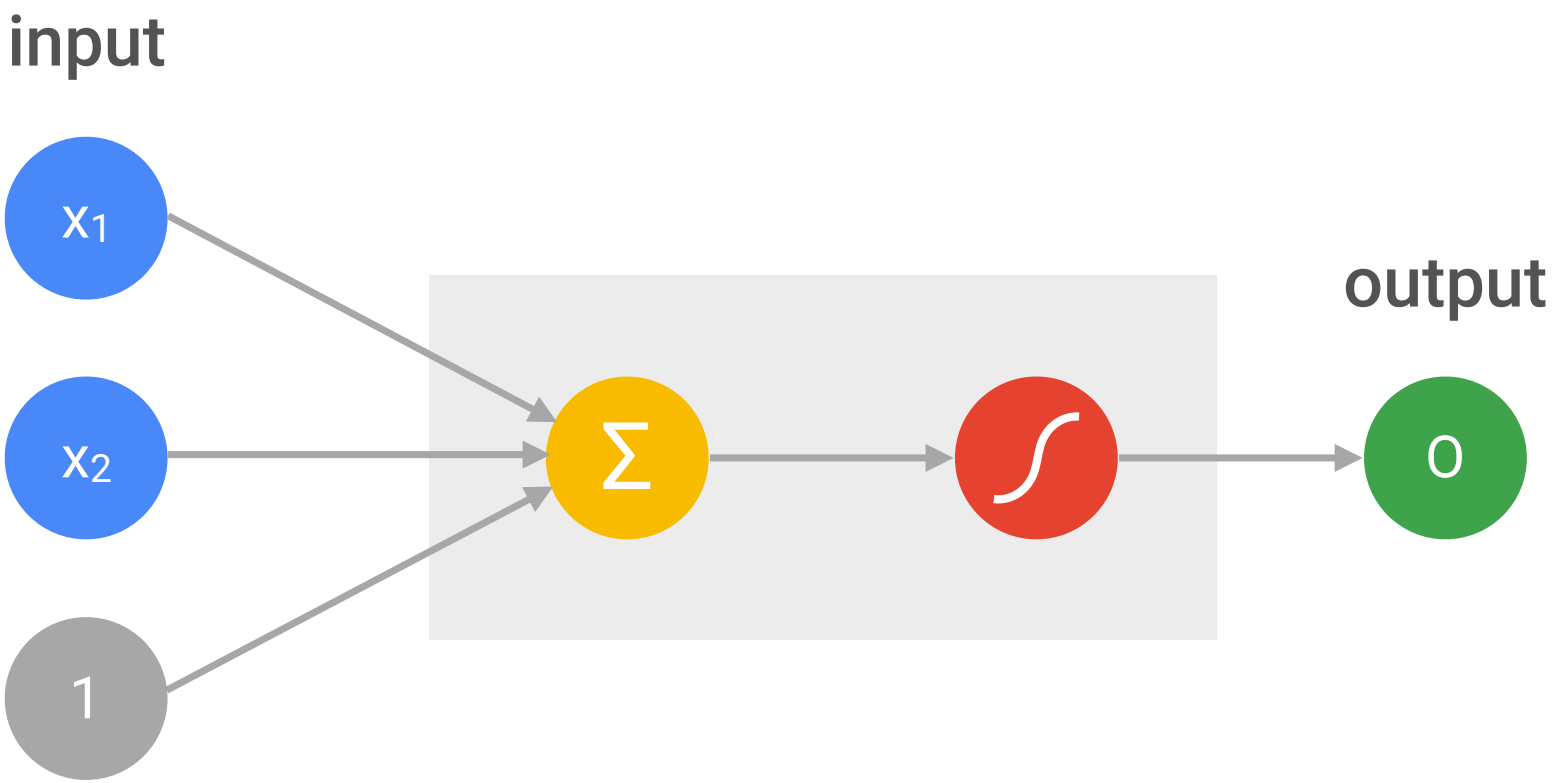


Source: Large-Scale Deep Learning for Intelligent Computer Systems, Jeff Dean, WSDM 2016

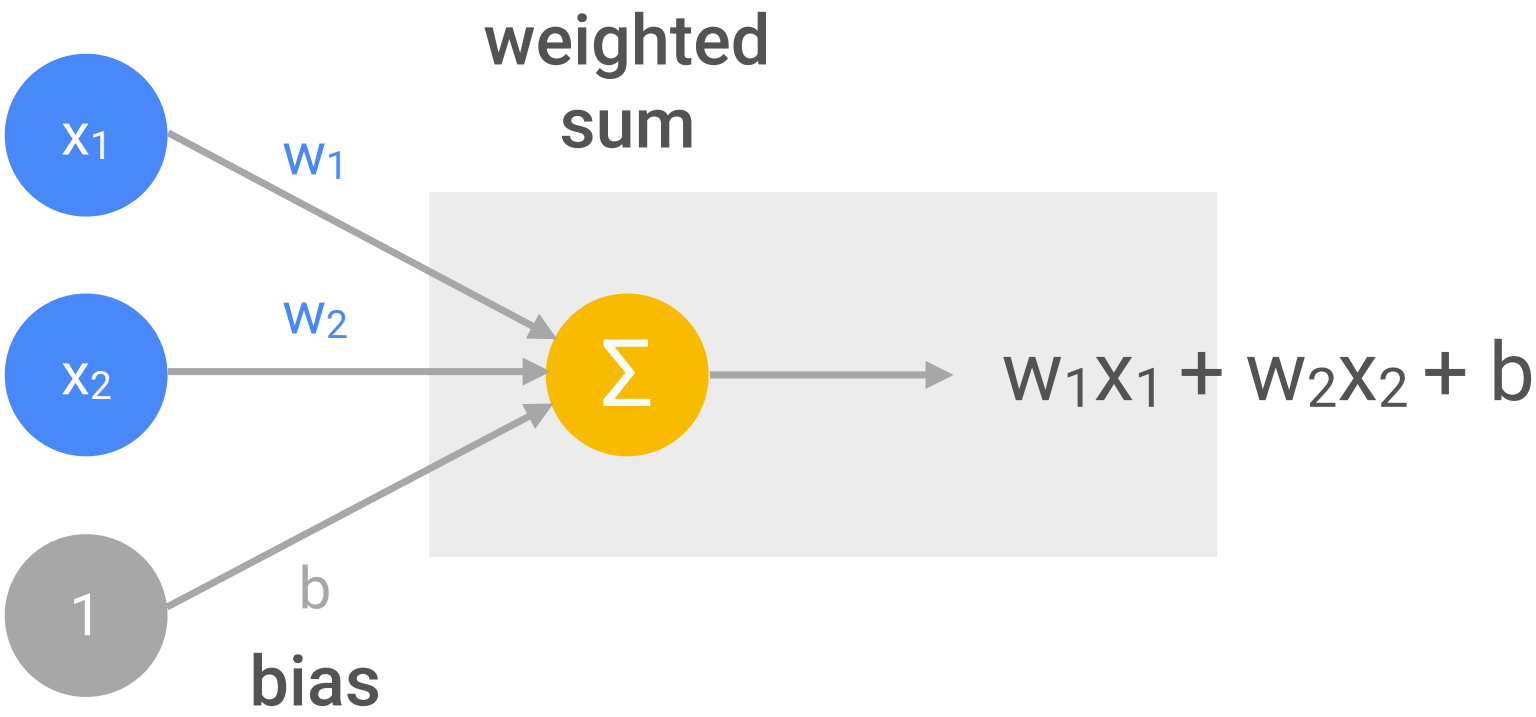

x_2
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
x_1
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
-1
0
1


x_2
6
5
4
3
2
1
0
-1
-2
-3
-4
-5
-6
x_1
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
-1
0
1

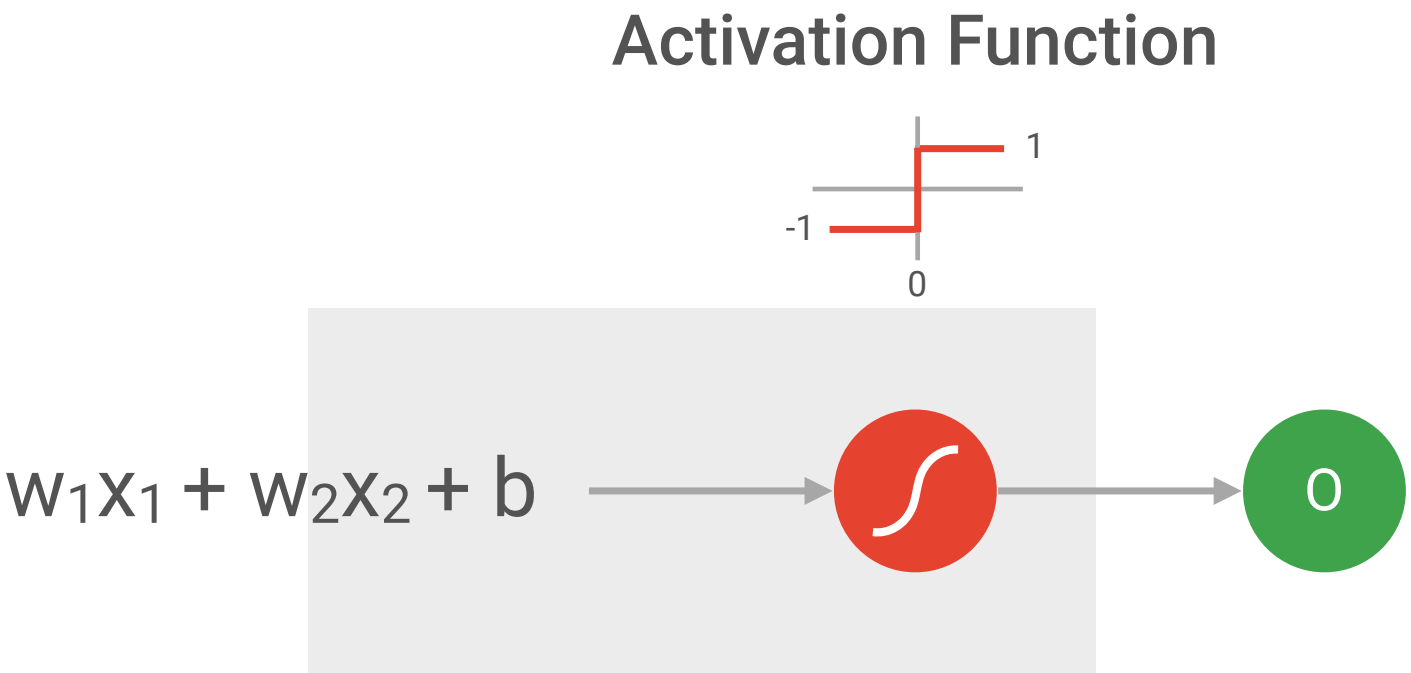
A Perceptron



A Perceptron



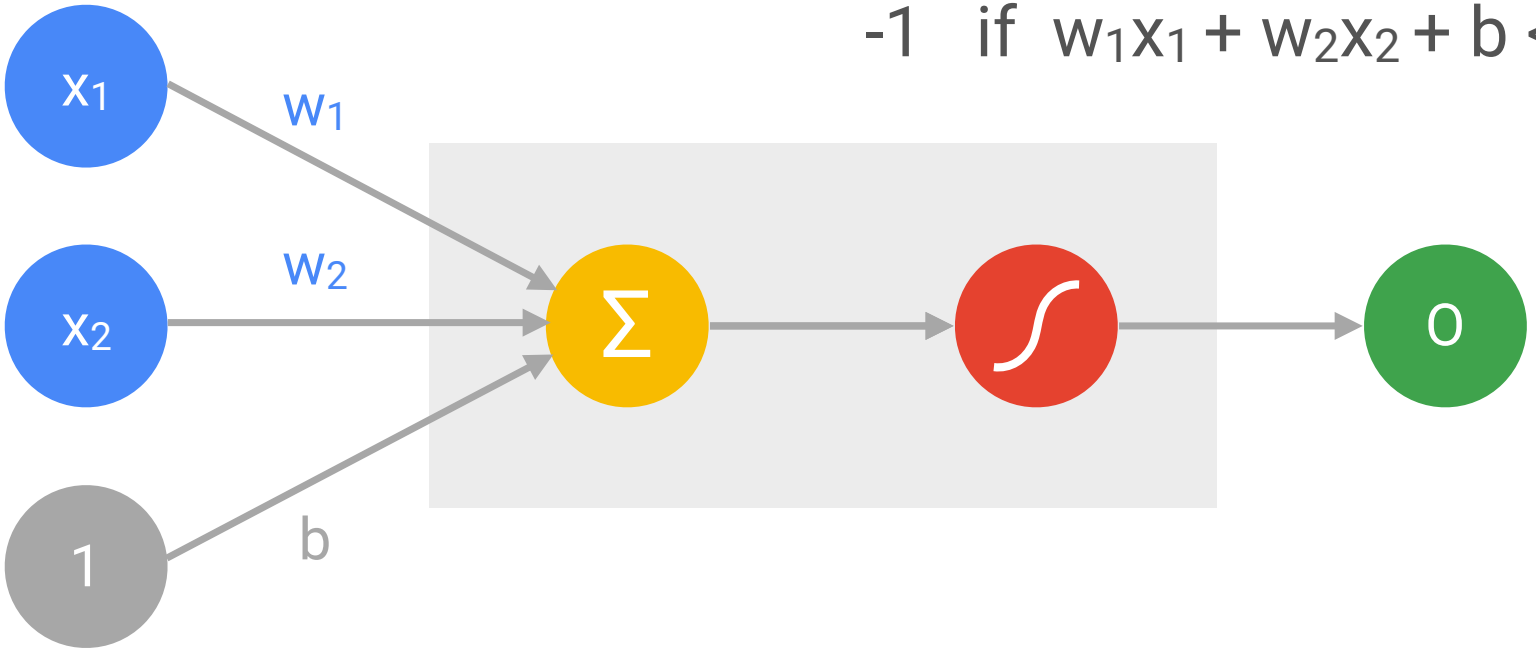
A Perceptron

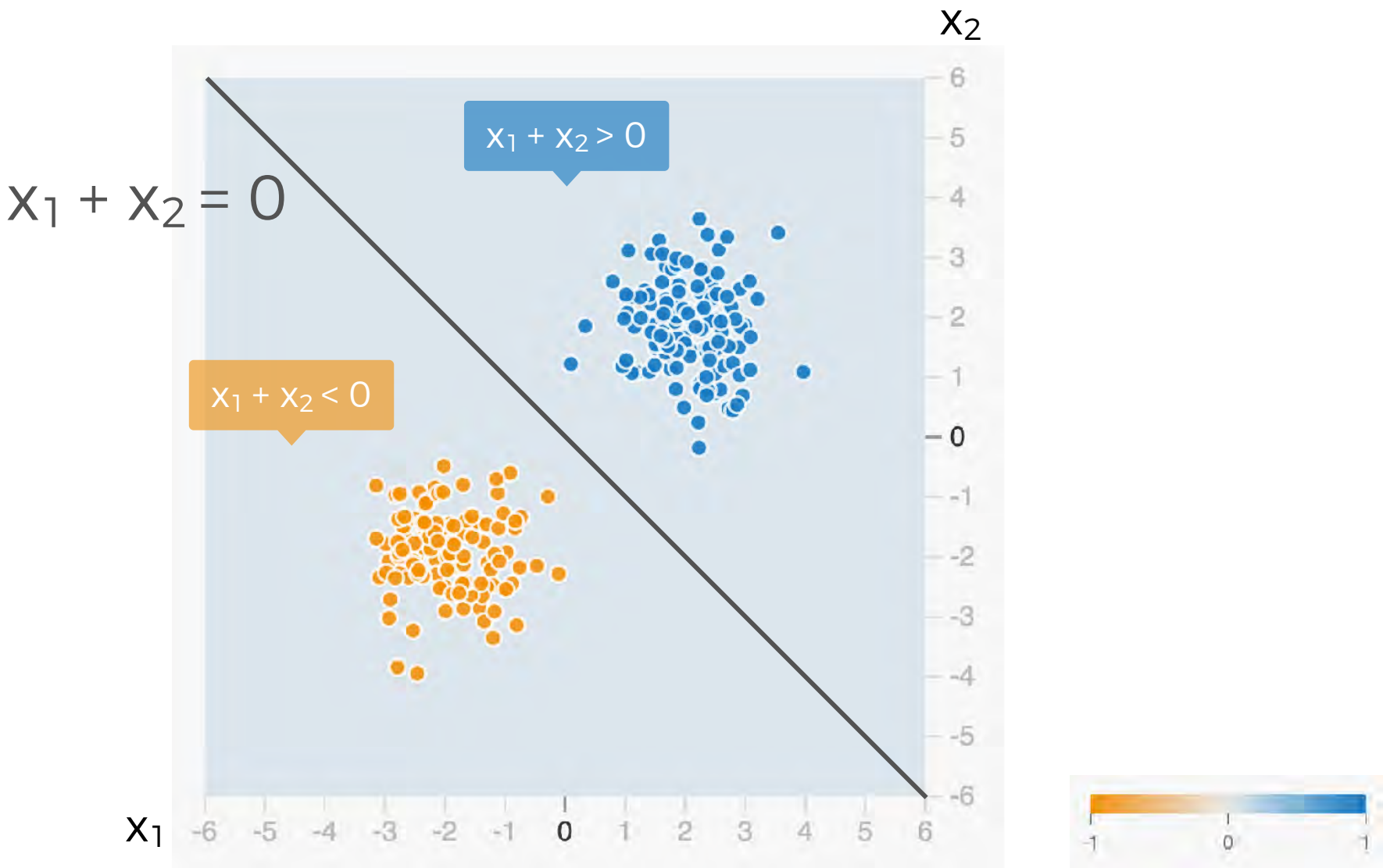


A Perceptron

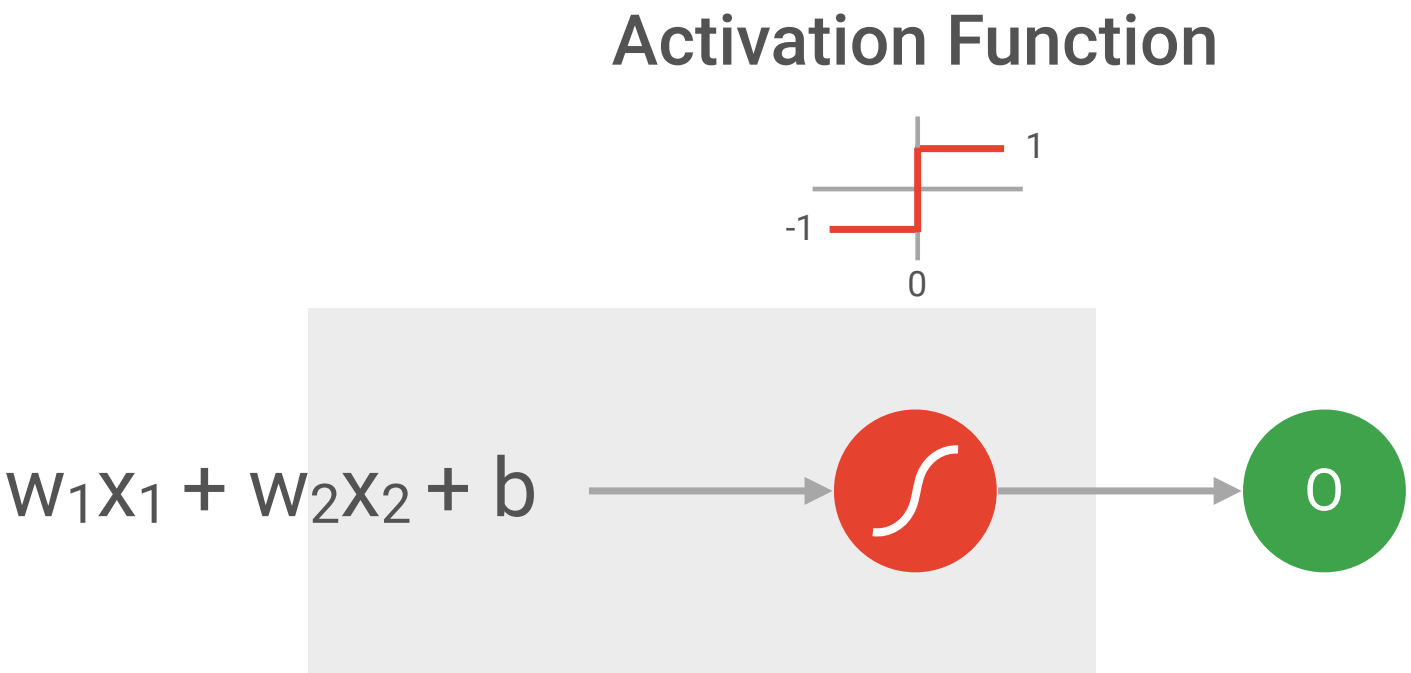
$$1 \quad \text{if} \quad w_1x_1 + w_2x_2 + b > 0$$

$$-1 \quad \text{if} \quad w_1x_1 + w_2x_2 + b < 0$$



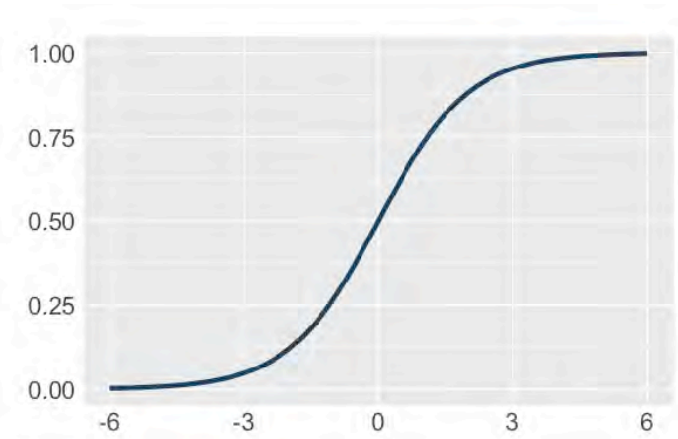

x_2
$x_1 + x_2 = 0$
$x_1 + x_2 > 0$
$x_1 + x_2 < 0$
x_1
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6
-1 0 1

A Perceptron



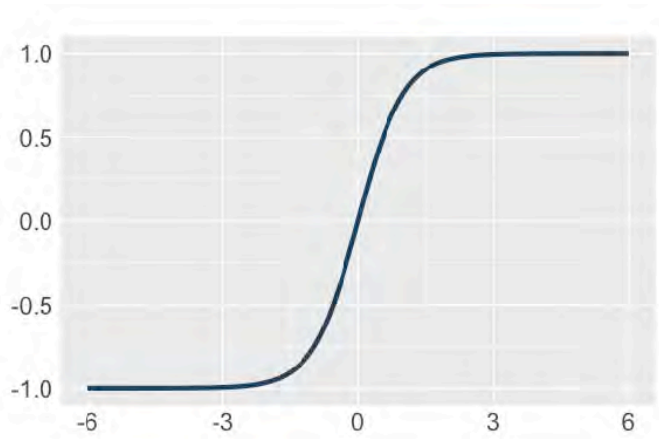
Other Activation Functions

Sigmoid



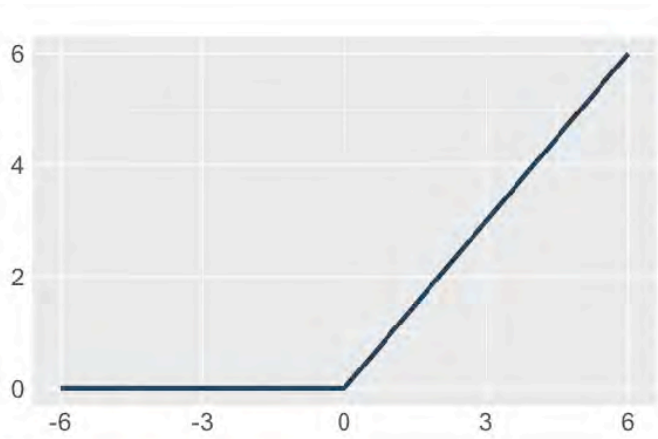
$$f(x) = \frac{1}{1 + e^{-x}}$$

TanH



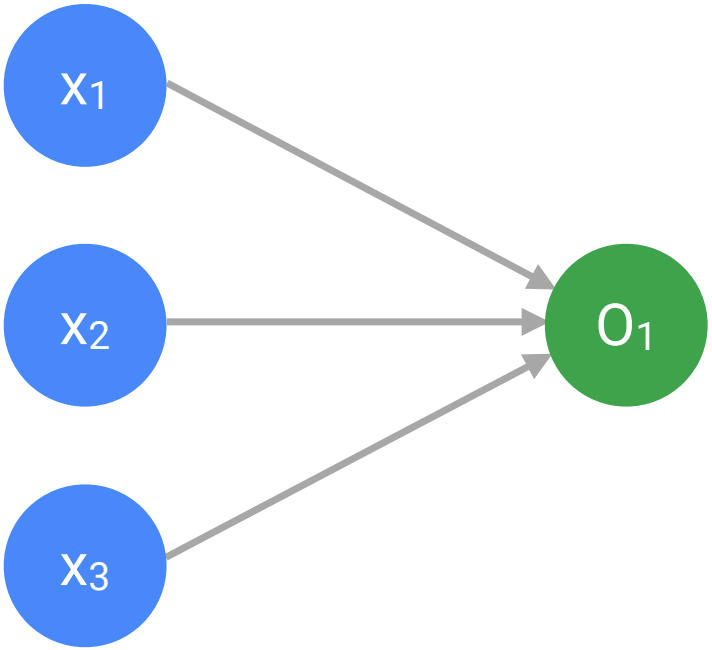
$$f(x) = \frac{2}{1 + e^{-2x}} - 1$$

ReLU

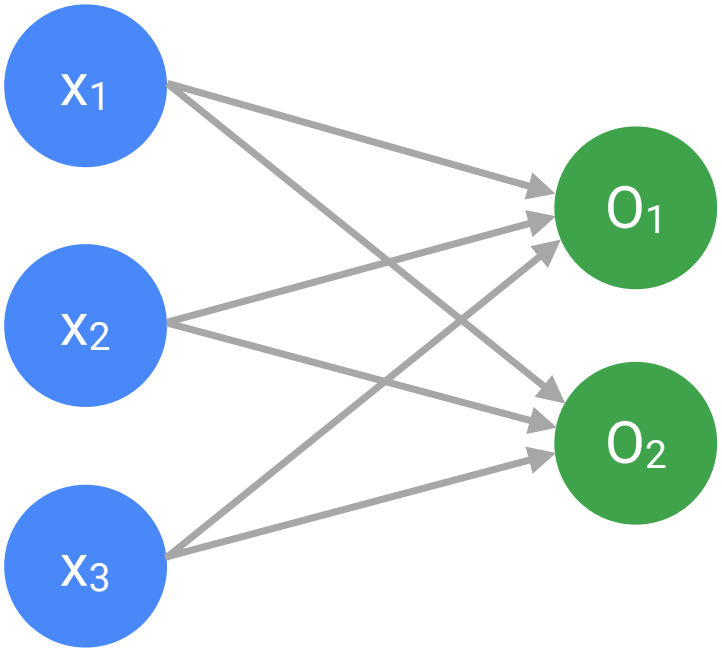


$$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ x & \text{for} \quad x \geq 0 \end{cases}$$

From a Perceptron

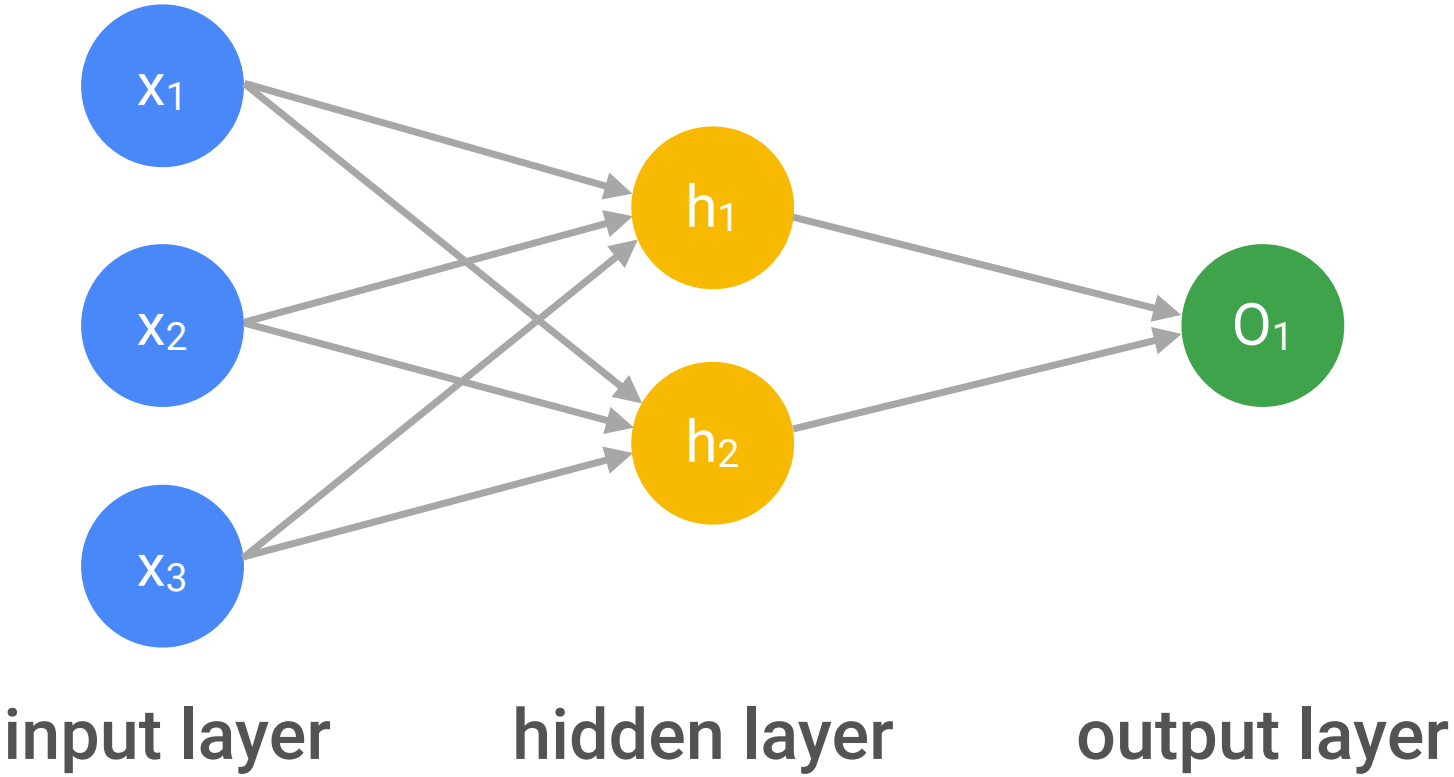


Multi-Output Perceptron

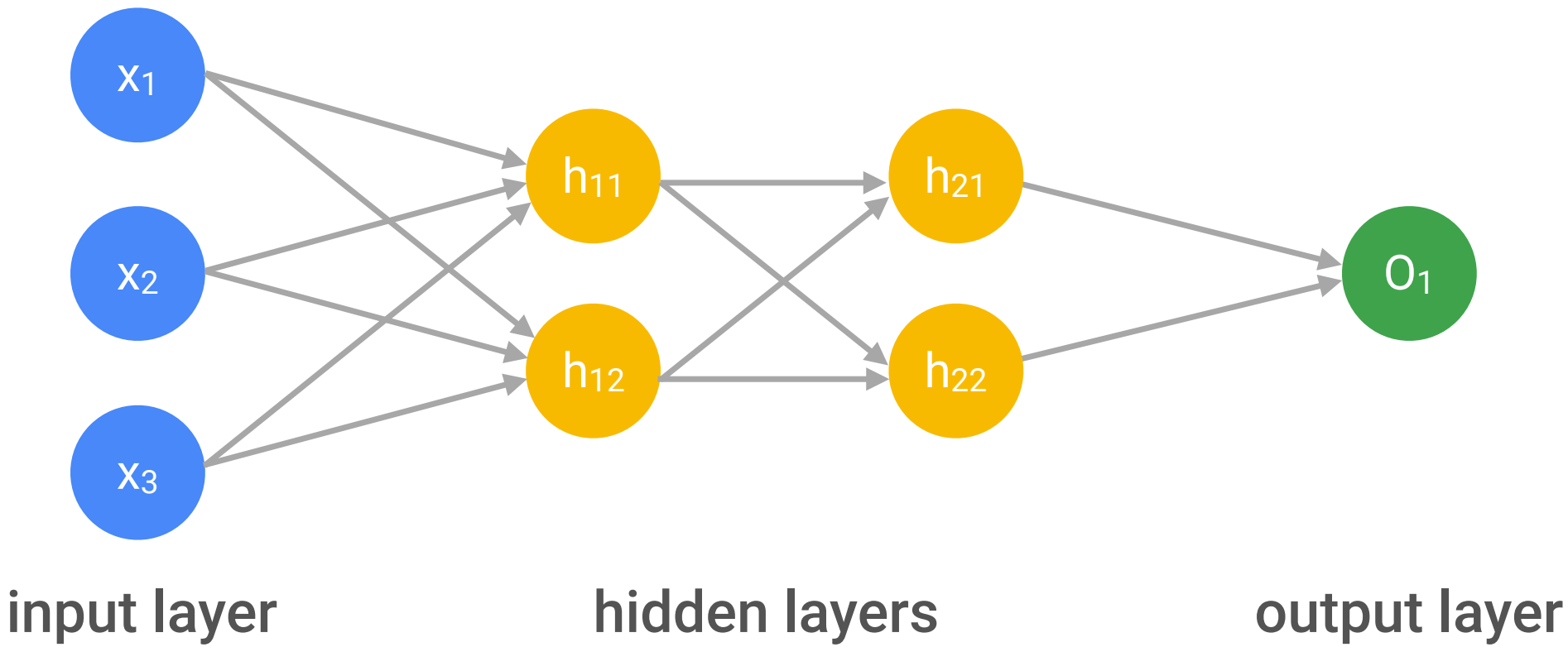


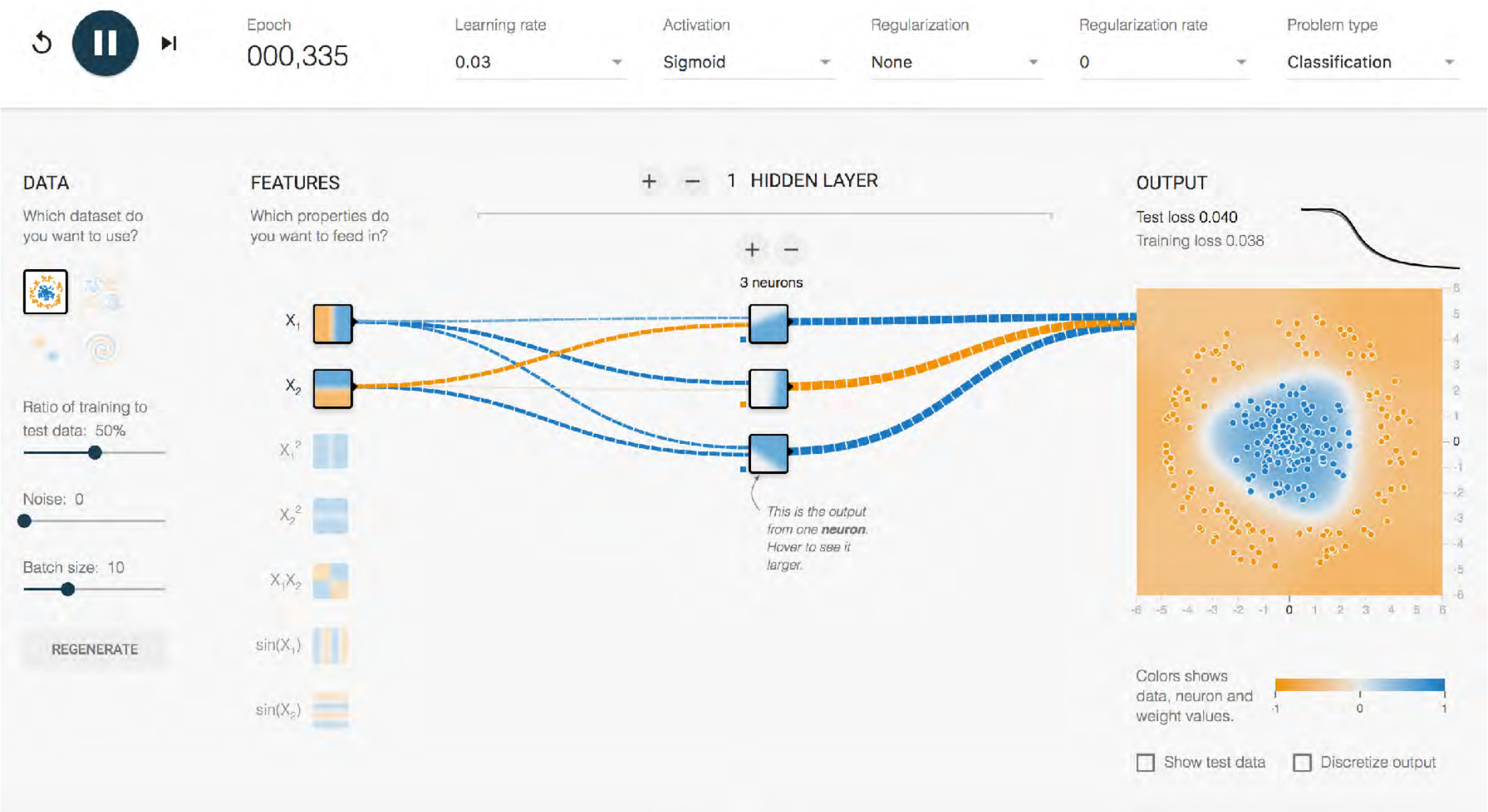
input layer output layer

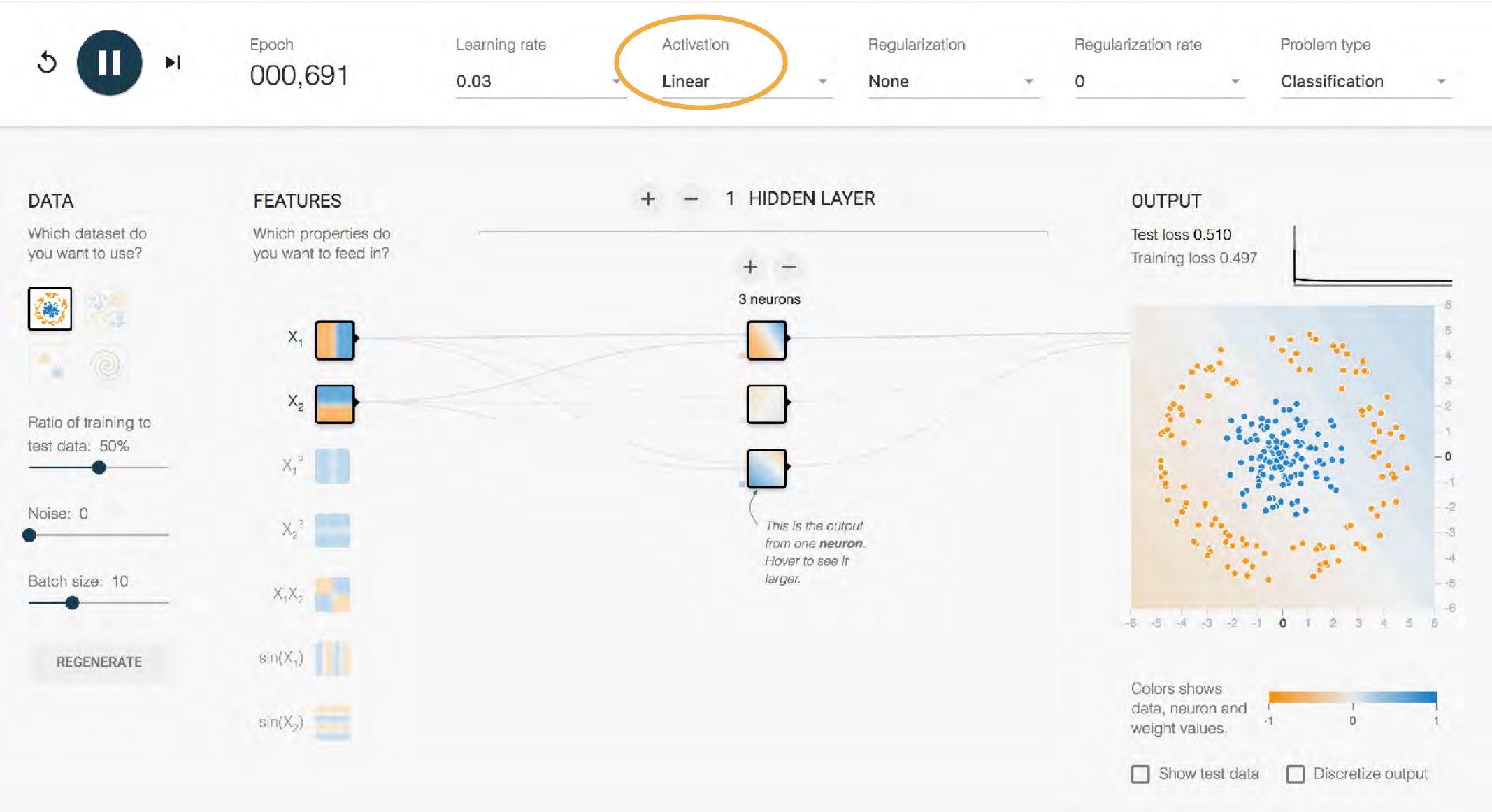
Multi-Layer Perceptron

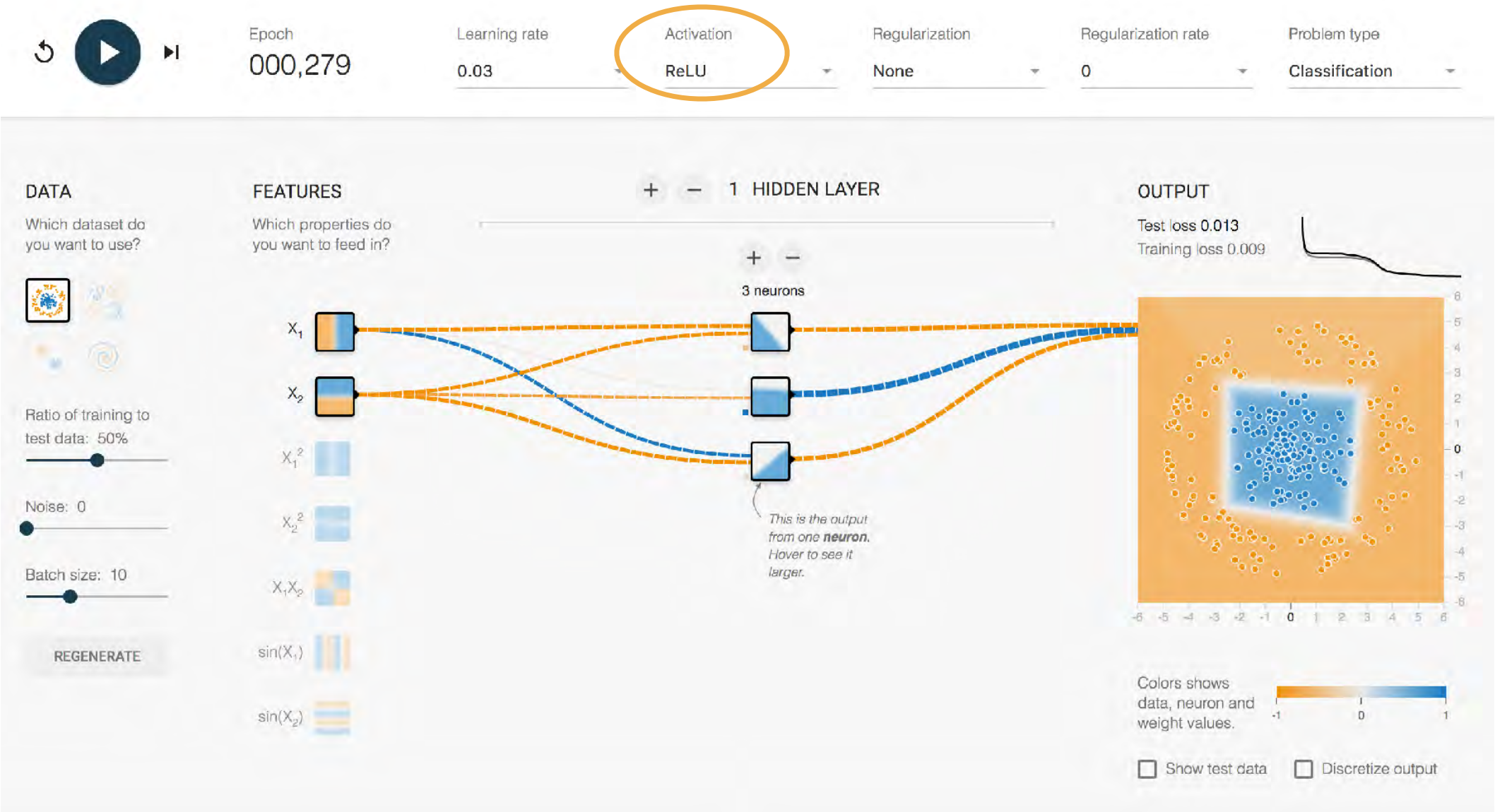


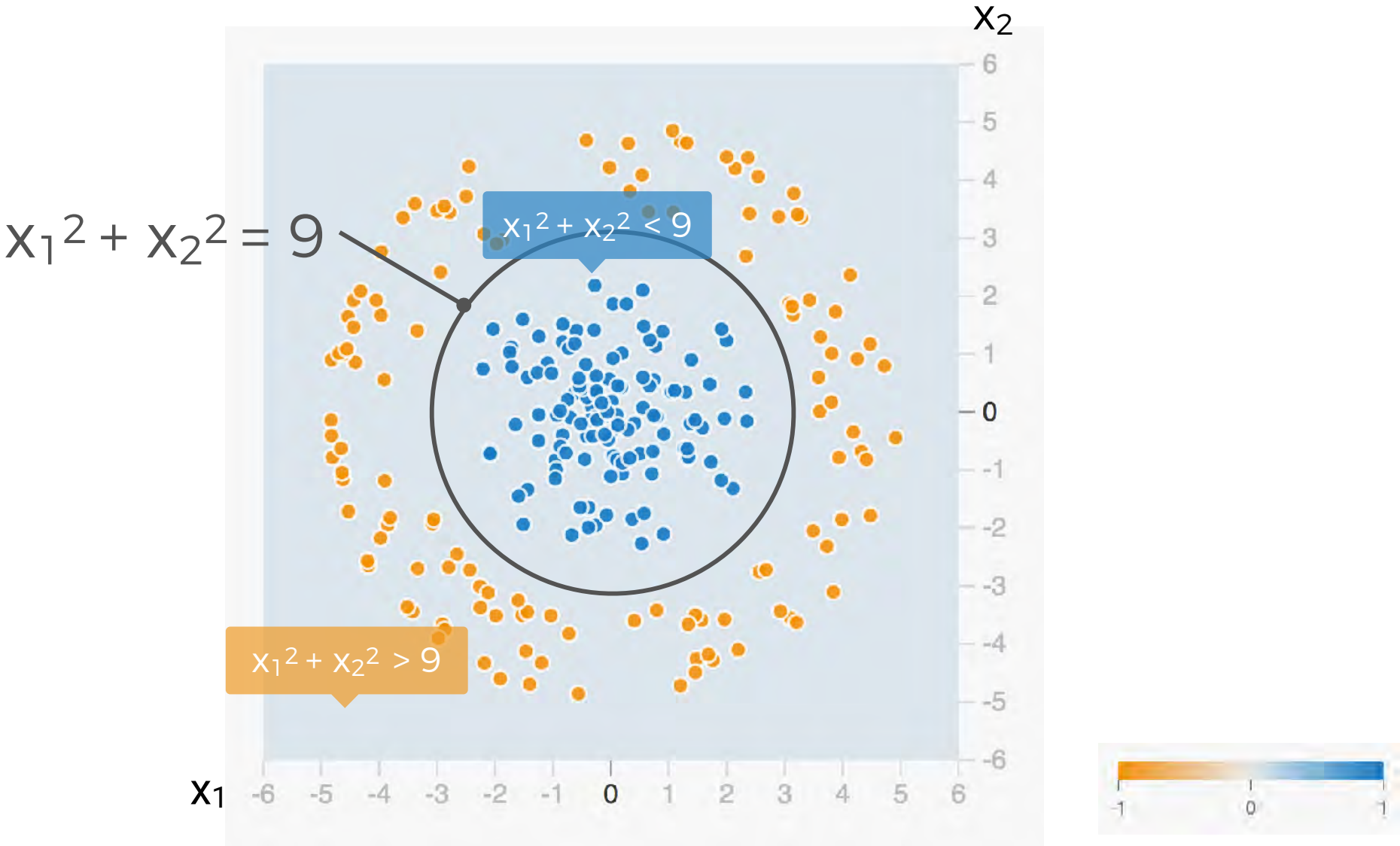
Deep Neural Network




Epoch
000,335
Learning rate
0.03
Activation
Sigmoid
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
1 HIDDEN LAYER
3 neurons
This is the output from one neuron. Hover to see it larger.
OUTPUT
Test loss 0.040
Training loss 0.038
Colors shows data, neuron and weight values.
Show test data
Discretize output


Epoch
000,691
Learning rate
0.03
Activation
Linear
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
1 HIDDEN LAYER
3 neurons
This is the output from one neuron. Hover to see it larger.
OUTPUT
Test loss 0.510
Training loss 0.497
Colors shows data, neuron and weight values.
Show test data
Discretize output


Epoch
000,279
Learning rate
0.03
Activation
ReLU
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
1 HIDDEN LAYER
3 neurons
This is the output from one neuron. Hover to see it larger.
OUTPUT
Test loss 0.013
Training loss 0.009
Colors shows data, neuron and weight values.
Show test data
Discretize output



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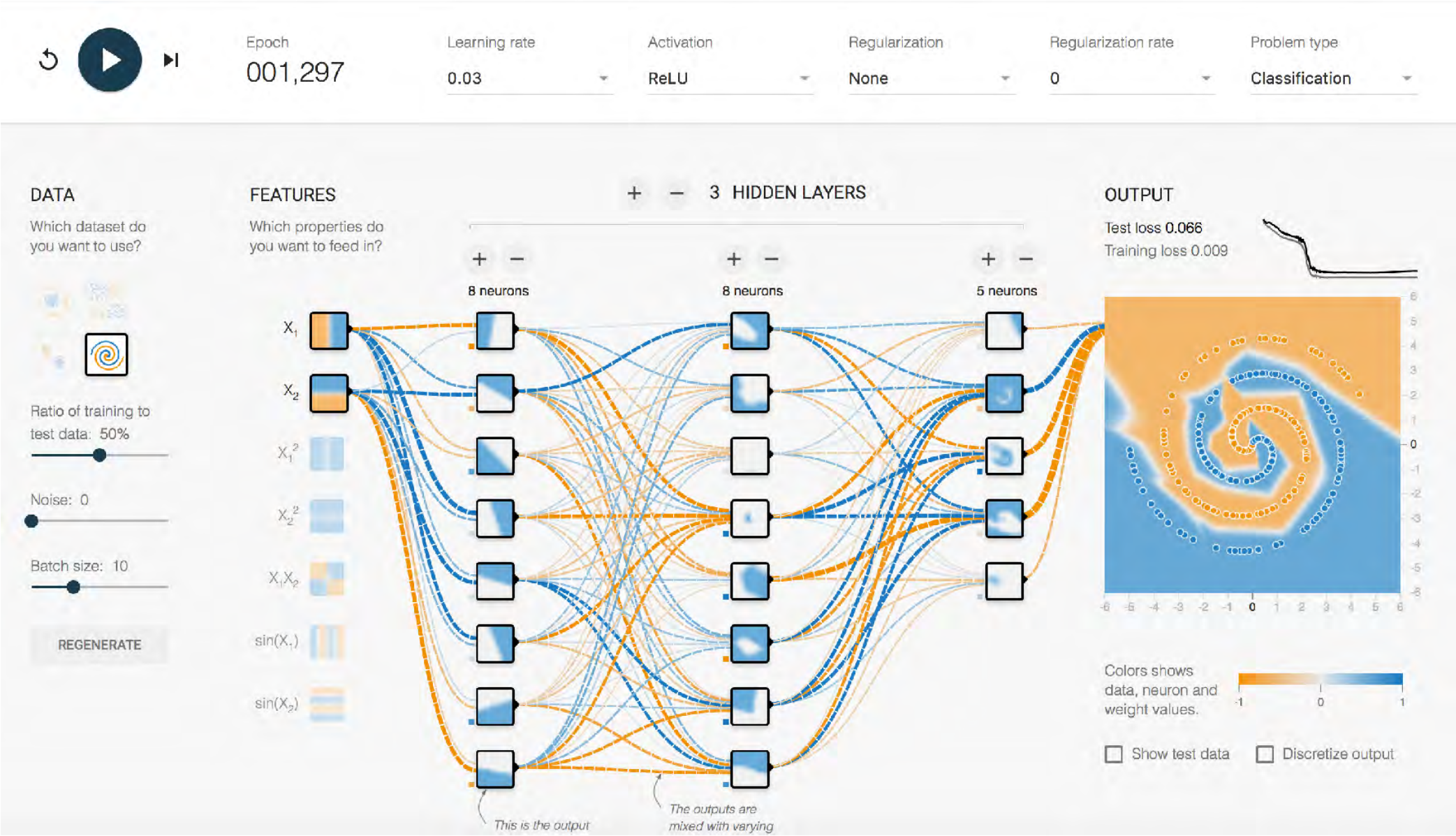
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
+ − 0 HIDDEN LAYERS
OUTPUT
Test loss 0.010
Training loss 0.009
Colors shows data, neuron and weight values.
-1 0 1
Show test data
Discretize output


Epoch
001,297
Learning rate
0.03
Activation
ReLU
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
3 HIDDEN LAYERS
8 neurons
8 neurons
5 neurons
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
This is the output
The outputs are mixed with varying
OUTPUT
Test loss 0.066
Training loss 0.009
Colors shows data, neuron and weight values.
Show test data
Discretize output

Why Deep Learning?

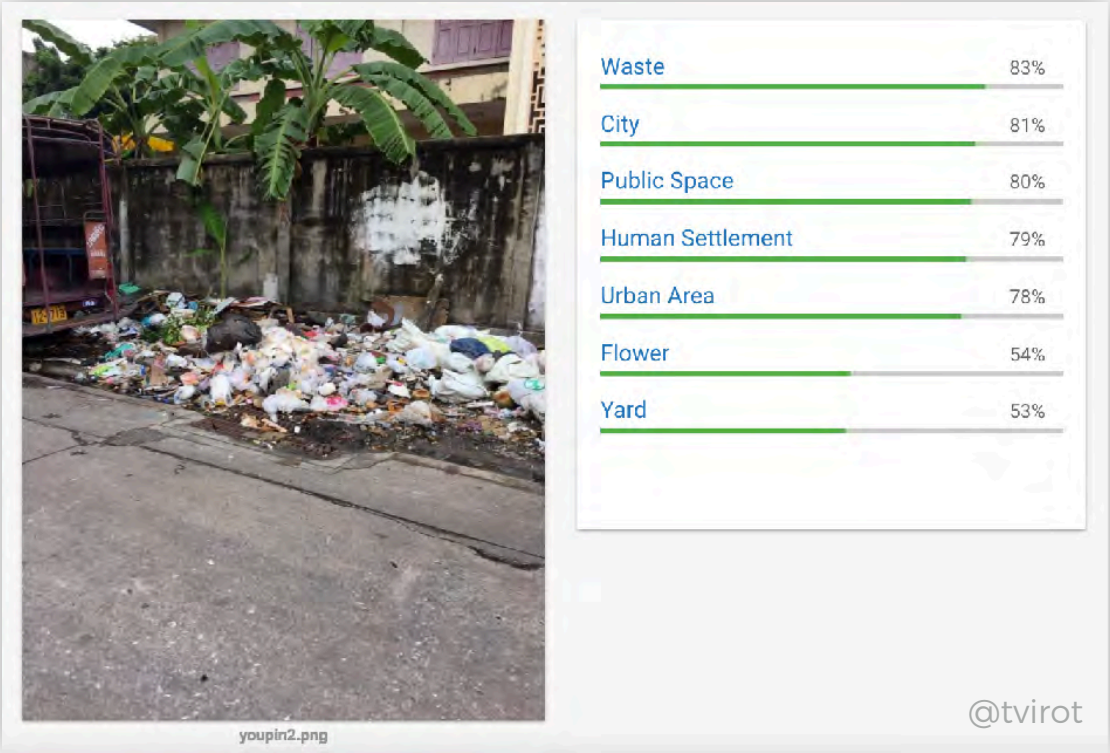
- Traditional machine learning requires **feature engineering**, and models may have **simplifying assumptions**
- Deep learning models are very flexible
- But...it requires a lot of data (and CPUs/GPUs)

Deep Learning for Images

Google Cloud Vision API

Understand the content of images

- Label Detection
- Optical Character Recognition
- Explicit Content Detection
- etc.

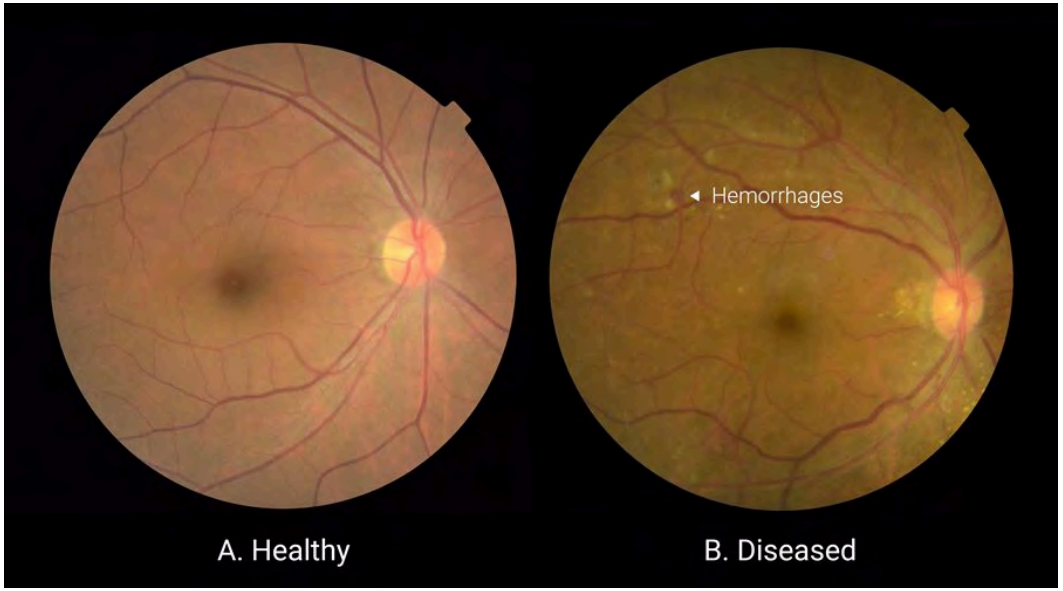


Source: https://youpin.city/app

Detecting Diabetic Eye Disease

A deep learning algorithm capable of interpreting signs of Diabetic Retinopathy (DR) in retinal photographs.

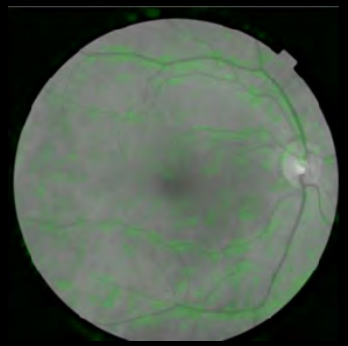
DR — an eye condition that affects people with diabetes — is the fastest growing cause of blindness, with nearly 415 million diabetic patients at risk worldwide.



Source: https://www.blog.google/topics/machine-learning/detecting-diabetic-eye-disease-machine-learning/

Image of Retina

Age



Actual: 57.6 years
Predicted: 59.1 years

Gender

Actual: female
Predicted: female

Smoking

Actual: non-smoker
Predicted: non-smoker

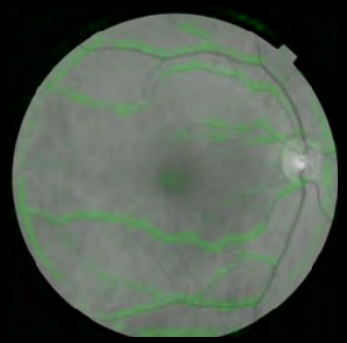
HbA1c

Actual: non-diabetic
Predicted: 6.7%

BMI

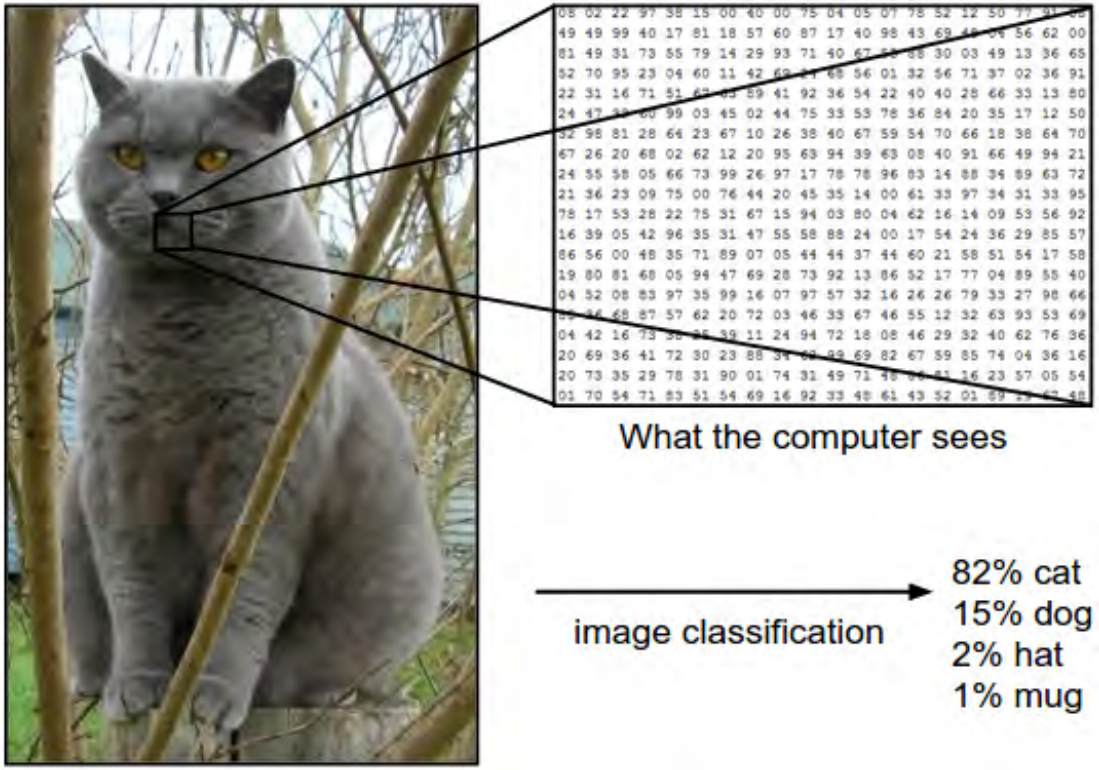
Actual: 26.3 kg m^{-2}
Predicted: 24.1 kg m^{-2}

SBP



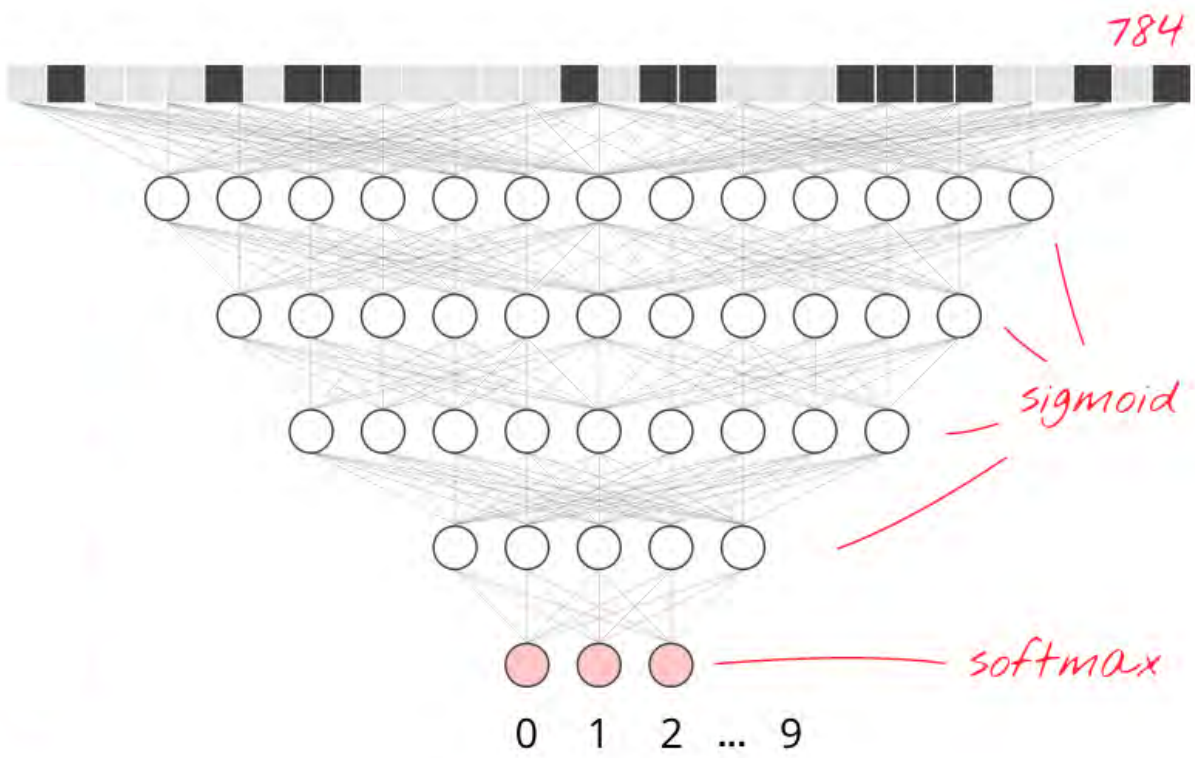
Actual: 148.5 mmHg
Predicted: 148.0 mmHg

Image Classification



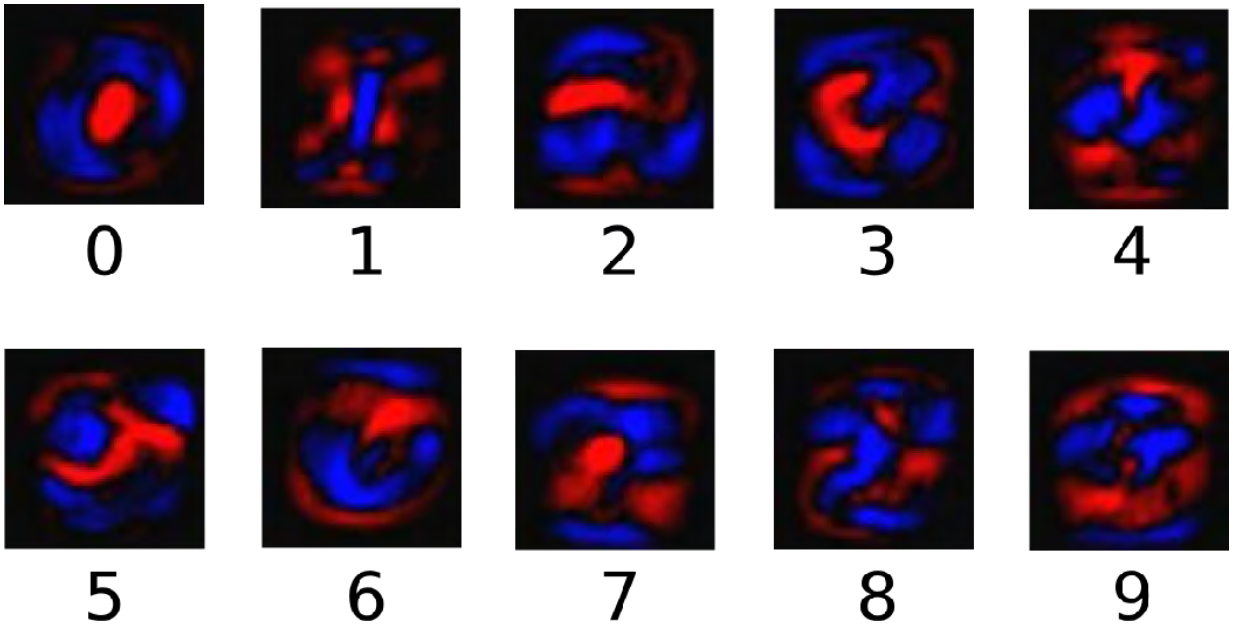
Source: http://cs231n.github.io/classification/

Fully-Connected Network



Source: https://codelabs.developers.google.com/codelabs/cloud-tensorflow-mnist/

Fully-Connected Network



Red represents negative weights, while blue represents positive weights.

Source: https://www.tensorflow.org/get_started/mnist/beginners

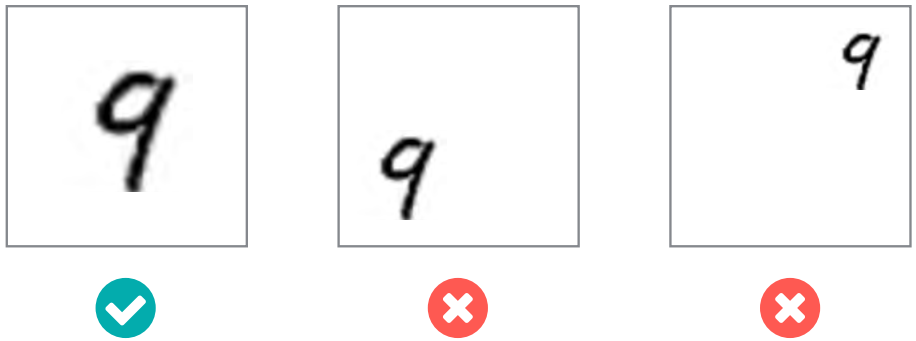
Challenges

Try this:
scs.ryerson.ca/~aharley/vis/fc

- Doesn't scale

  e.g., 300 x 300 RGB image would require 270,000 weights **for each neuron** in the first hidden layer of the neural network

- Easily overfitted



Source: https://codelabs.developers.google.com/codelabs/cloud-tensorflow-mnist/

Convolutional Neural Networks (CNN)

Neural networks that share their parameters across space.

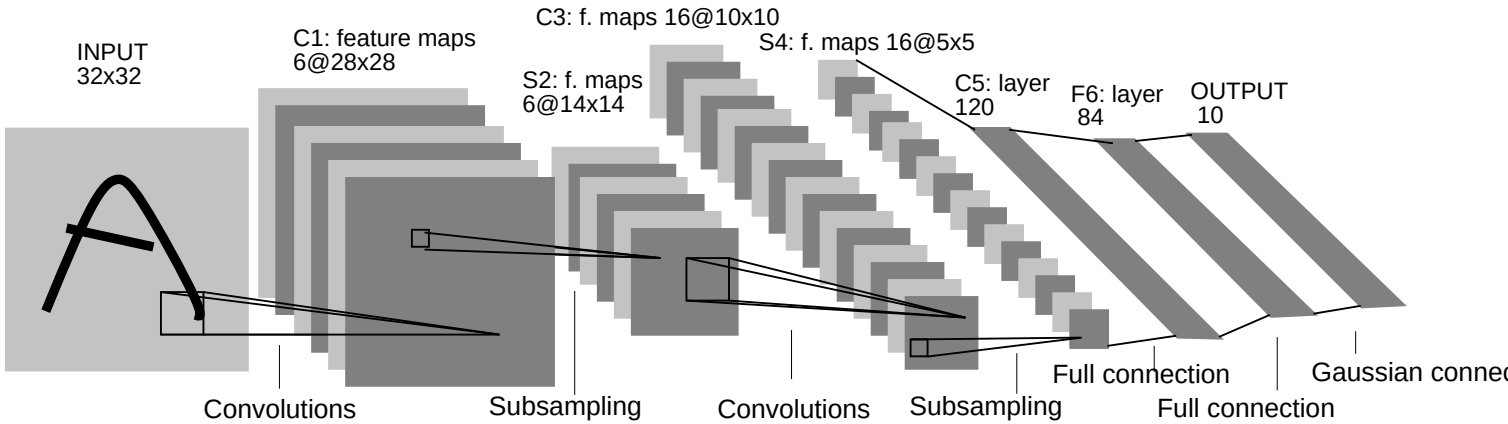
CNNs / ConvNets

Designed to emulate the behavior of a visual cortex

- Assume that the inputs are images
- Apply **filters** to one small region at a time to detect certain features of an object (edge, circle, or certain shapes)
- A feature of an object is **translation invariant**, and the filters applied to each small region can **share weights/ parameters**
- Fewer parameters required. More robust.

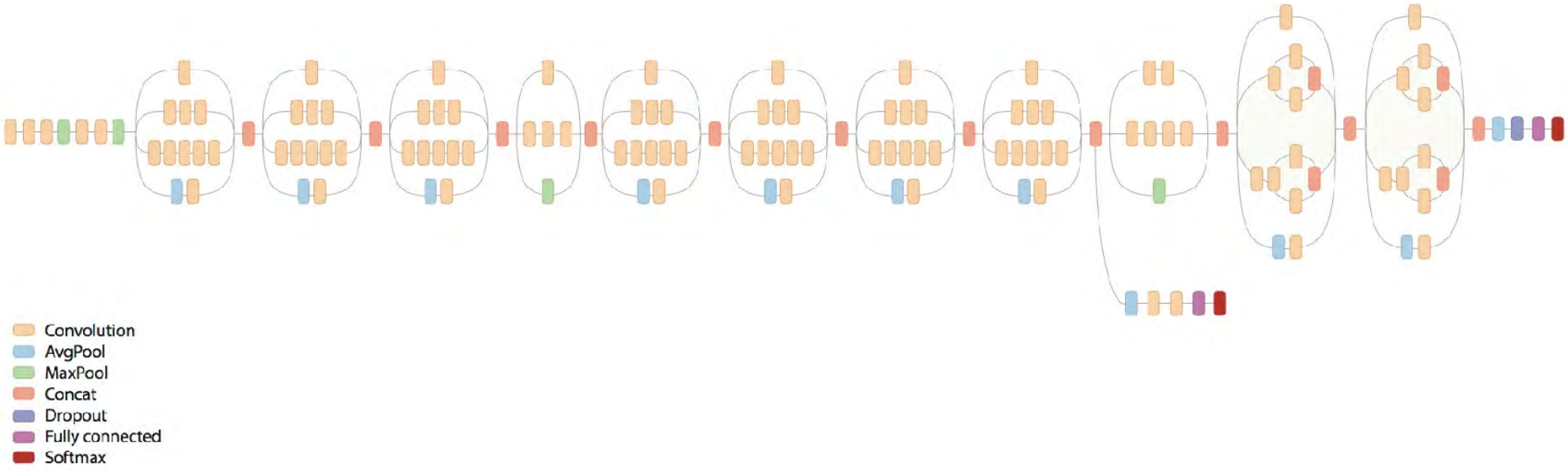
Layers in CNNs

1. Convolutional Layer
2. Pooling Layer
3. Fully Connected Layer

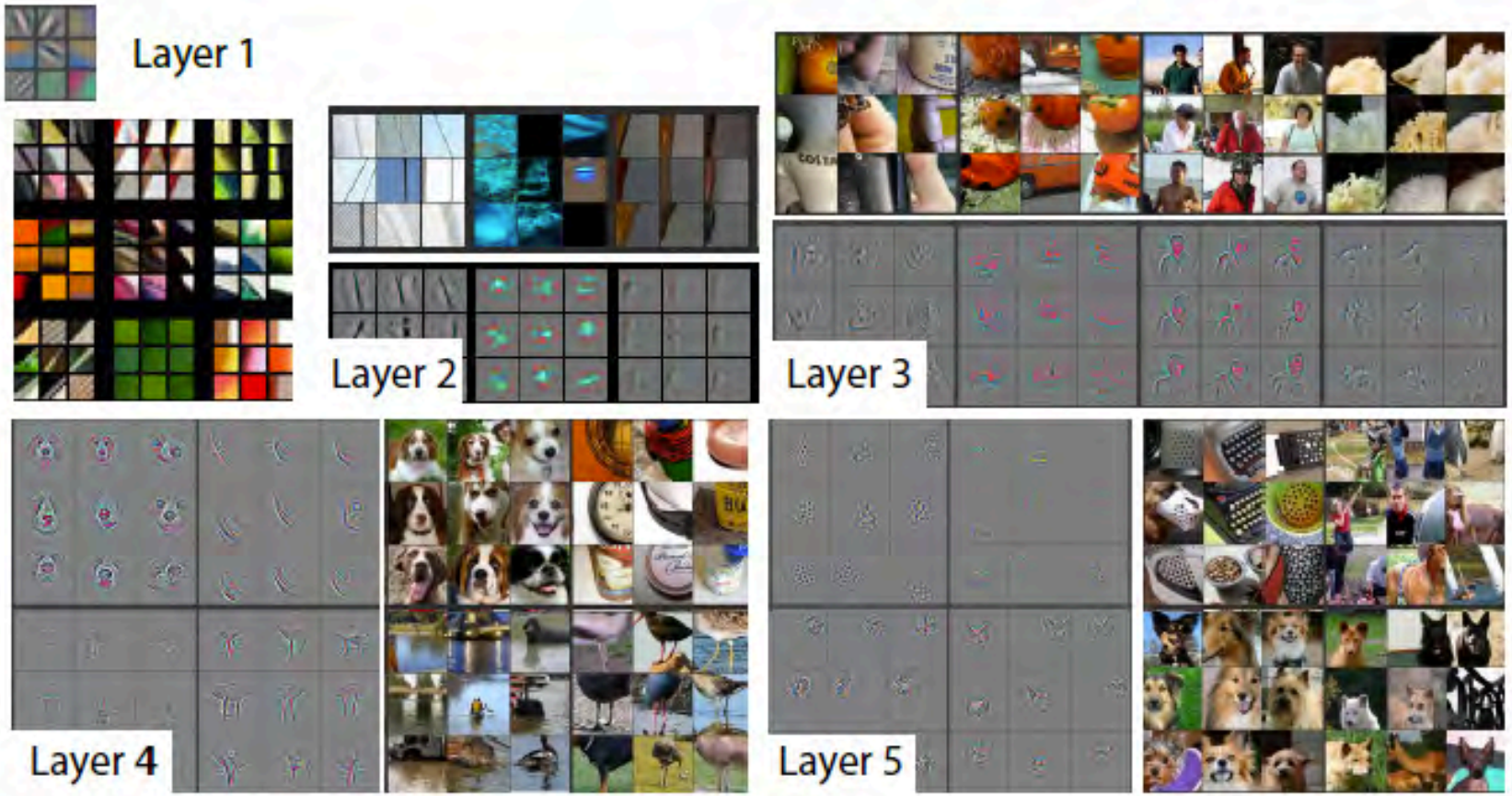


Source: Y. Lecun, L. Bottou, Y. Bengio and P. Haffner, "Gradient-based learning applied to document recognition," in Proceedings of the IEEE, vol. 86, no. 11, pp. 2278-2324, Nov 1998.

Inception V3



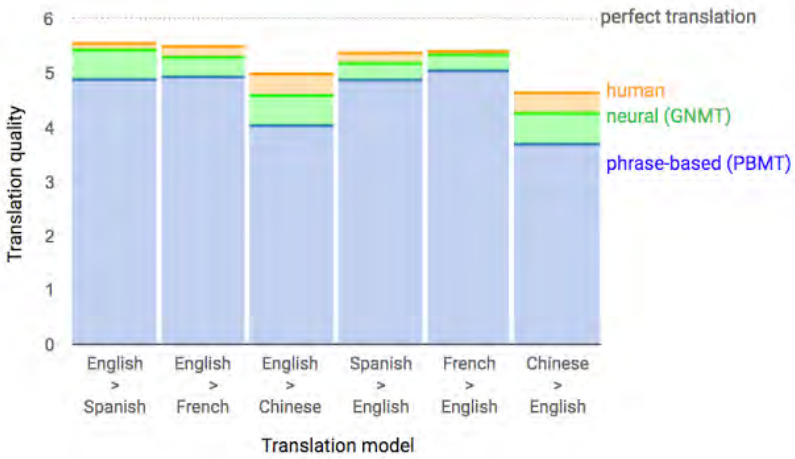
Source: Google Codelabs

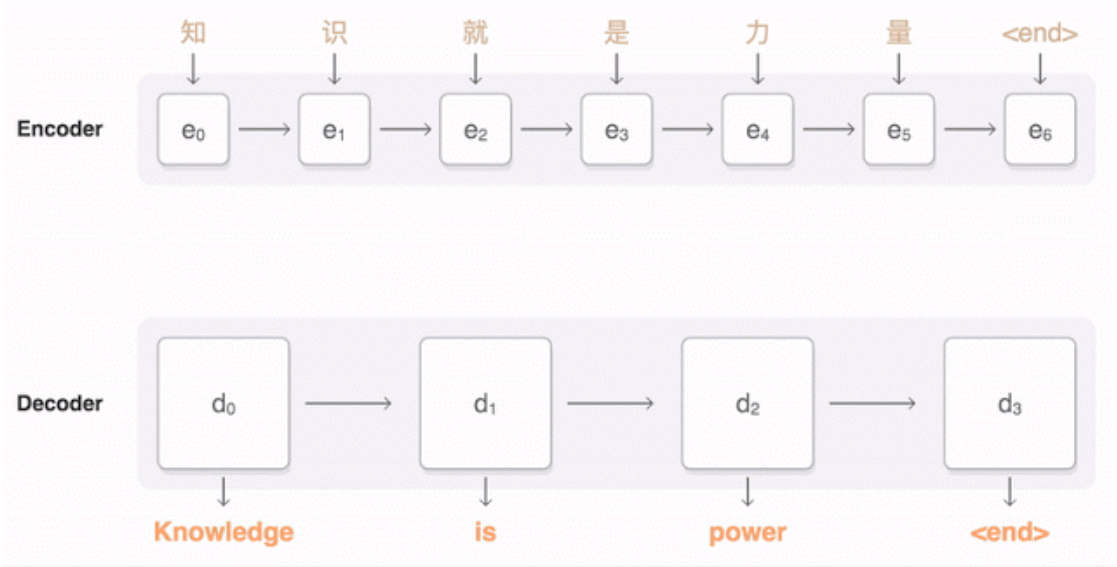


Source: Visualizing and Understanding Convolutional Networks (Zeiler and Fergus, 2014)

Deep Learning for Text

Machine Translation





Sources: https://research.googleblog.com/2016/09/a-neural-network-for-machine.html | https://cloud.google.com/translate/

Text Summarization

A deep reinforced model for abstractive summarization

cia documents reveal iot-specific televisions can be used to secretly record conversations .
cybercriminals who initiated the attack managed to commandeer a large number of internet-connected devices in current use .
cia documents revealed that microwave ovens can spy on you - maybe if you personally don't suffer the consequences of the sub-par security of the iot .

Internet of Things (IoT) security breaches have been dominating the headlines lately . WikiLeaks's trove of CIA documents revealed that internet-connected televisions can be used to secretly record conversations . Trump's advisor Kellyanne Conway believes that microwave ovens can spy on you - maybe she was referring to microwave cameras which indeed can be used for surveillance . And don't delude yourself that you are immune to IoT attacks , with 96 % of security professionals responding to a new survey expecting an increase in IoT breaches this year . Even if you personally don't suffer the consequences of the sub-par security of the IoT , your connected gadgets may well be unwittingly cooperating with criminals . Last October , Internet service provider Dyn came under an attack that disrupted access to popular websites . The cybercriminals who initiated the attack managed to commandeer a large number of internet-connected devices (mostly DVRs and cameras) to serve as their helpers . As a result , cybersecurity expert Bruce Schneier has called for government regulation of the IoT , concluding that both IoT manufacturers and their customers don't care about the security of the 8.4 billion internet-connected devices in current use . Whether because of government regulation or good old-fashioned self-interest , we can expect increased investment in IoT security technologies . In its recently-released TechRadar report for security and risk professionals , Forrester Research discusses the outlook for the 13 most relevant and important IoT security technologies , warning that " there is no single , magic security bullet that can easily fix all IoT security issues . " Based on Forrester's analysis , here's my list of the 6 hottest technologies for IoT security : IoT network security : Protecting and securing the network connecting IoT devices to back-end systems on the internet . IoT network security is a bit more challenging than traditional network security because there is a wider range of communication protocols , standards , and device capabilities , all of which pose significant issues and increased complexity . Key capabilities include traditional endpoint security features such as antivirus and antimalware as well as other features such as firewalls and intrusion prevention and detection systems . Sample vendors : Bayshore Networks , Cisco , Darktrace , and Senrio . IoT authentication : Providing the ability for users to authenticate an IoT device , including managing multiple users of a single device (such as a connected car) , ranging from simple static password/pins to more robust authentication mechanisms such as two-factor

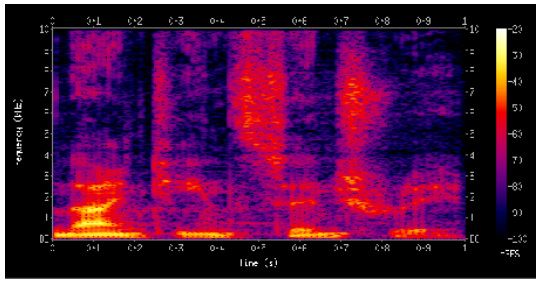
Source: https://einstein.ai/research/your-tldr-by-an-ai-a-deep-reinforced-model-for-abstractive-summarization

Sequence Models

Sequence Data

Data points are related or dependent on each other, e.g.,

- Text — each word in a sentence is dependent on other word in the sentence
- Audio waves
- Videos



Source: Wikimedia Commons

Word Order Matters

- Same set of words, different order.

The talk was **good**, not **bad** at all.

VS.

The talk was **bad**, not **good** at all.

Adapted from MIT's 6.S191: Introduction to Deep Learning (Lecture 2)

Statistical Invariance

- Same meaning regardless of position

Last month, I attended a Deep Learning workshop.

VS.

I attended a Deep Learning workshop **last month.**

Adapted from MIT's 6.S191: Introduction to Deep Learning (Lecture 2)

Long-Term Dependencies

- Information from the far past matters

In **Bangkok**, I had a great time and learnt to speak **Thai** a little bit.

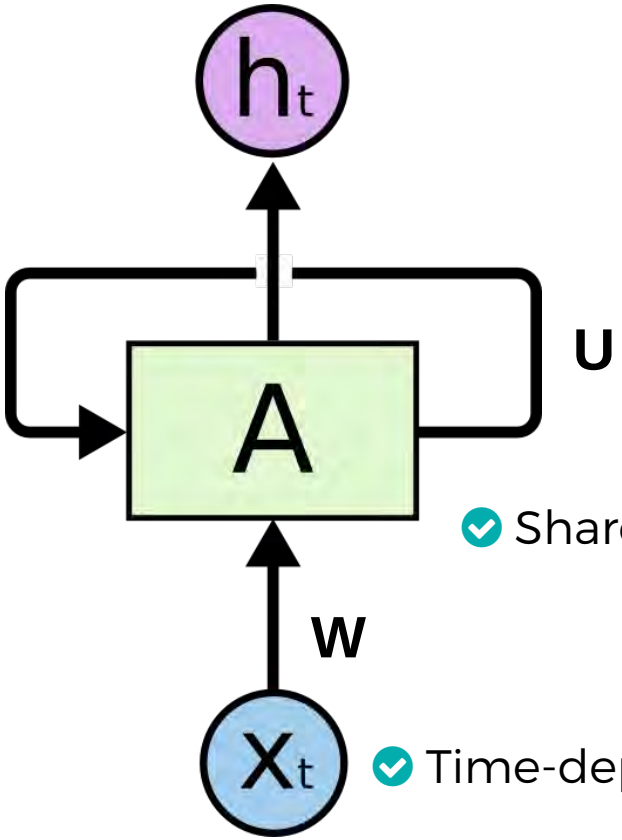
Adapted from MIT's 6.S191: Introduction to Deep Learning (Lecture 2)

Recurrent Neural Networks (RNN)

RNNs

Recurrent Neural Networks
have loops

Capture past information

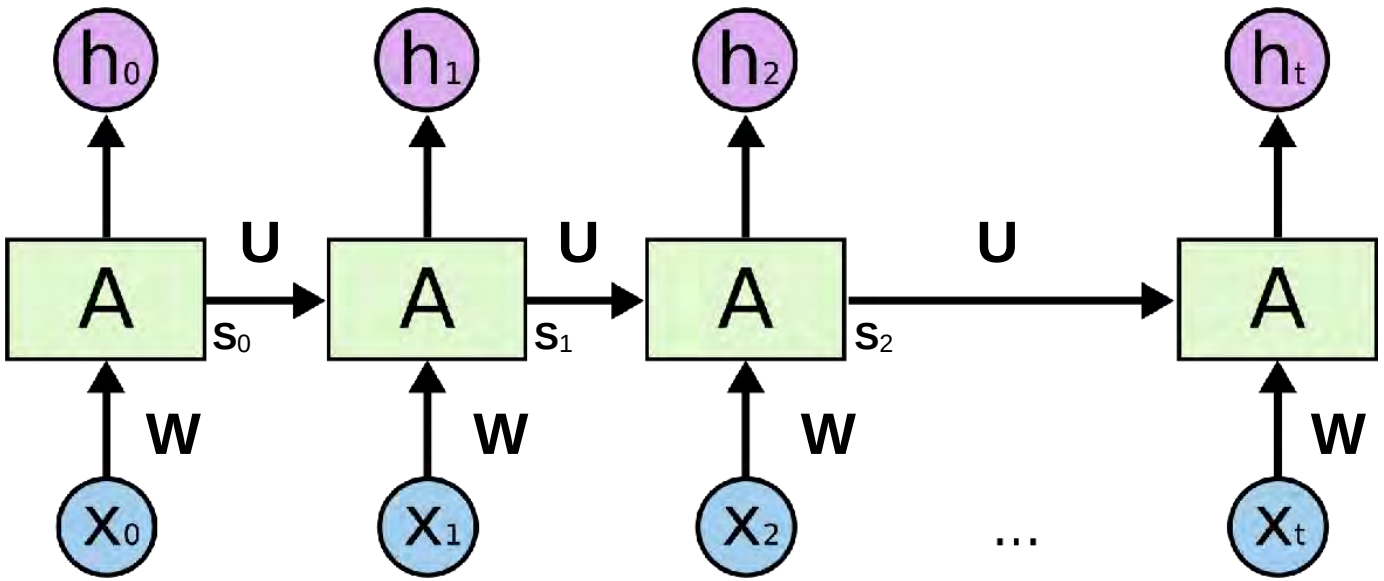


Share parameters

Time-dependent

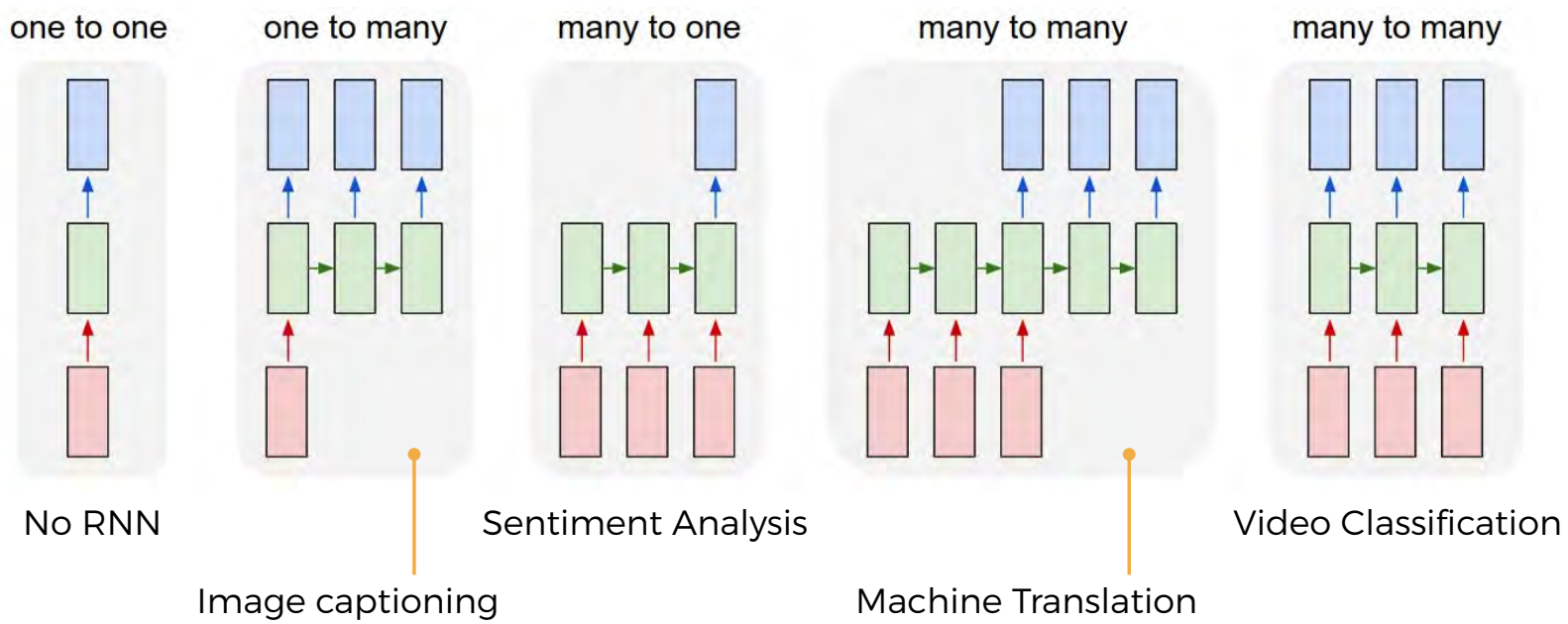
Source: http://colah.github.io/posts/2015-08-Understanding-LSTMs/

Unfolded RNN



Source: http://colah.github.io/posts/2015-08-Understanding-LSTMs/

RNN Architectures



Source: http://karpathy.github.io/2015/05/21/rnn-effectiveness/

APPLICATIONS IN ECONOMICS

Policy and Decision Problems

Where not to eat?

Health Inspection

90

out of 100

Violations

- Moderate risk food holding temperature [date violation corrected: 11/10/2016]
- Improper storage of equipment utensils or linens [date violation corrected: 11/10/2016]
- Inadequate and inaccessible handwashing facilities [date violation corrected: 11/10/2016]

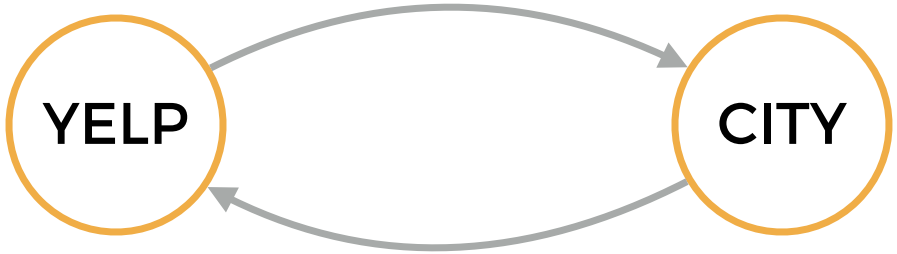
Inspections

Date	Inspection Type	Violations	Score
November 3, 2016	Routine	3	90
February 1, 2016	Routine	2	94
February 3, 2015	Routine	2	94
May 7, 2014	Routine	2	92

Source: Yelp

Where not to eat?

Predicting hygiene score based on ratings and reviews to help health departments better utilize their resources



Sharing inspection data to be displayed alongside Yelp reviews

Source: https://hbr.org/2015/02/city-governments-are-using-yelp-to-tell-you-where-not-to-eat

New Data Sources

Detecting influenza epidemics using search engine query data

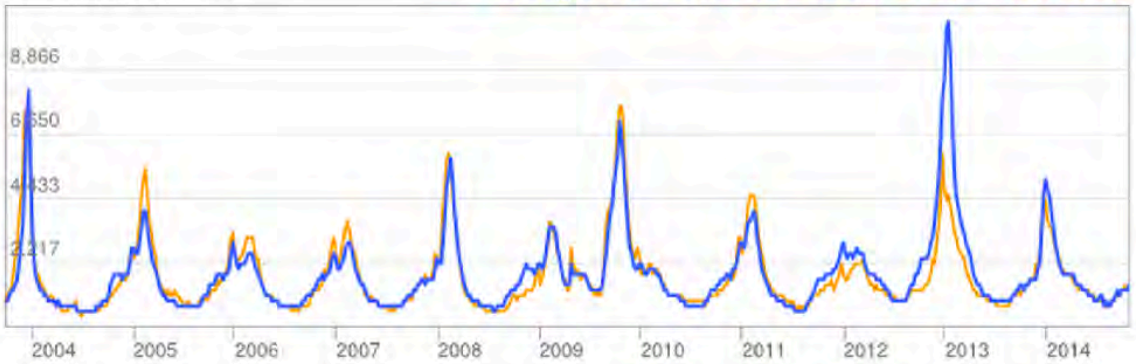
Jeremy Ginsberg[1], Matthew H. Mohebbi[1], Rajan S. Patel[1], Lynnette Brammer[2], Mark S. Smolinski[1] & Larry Brilliant[1]

[1]Google Inc. [2]Centers for Disease Control and Prevention

Historical estimates See data for: United States

United States Flu Activity

Influenza estimate ● Google Flu Trends estimate ● United States data

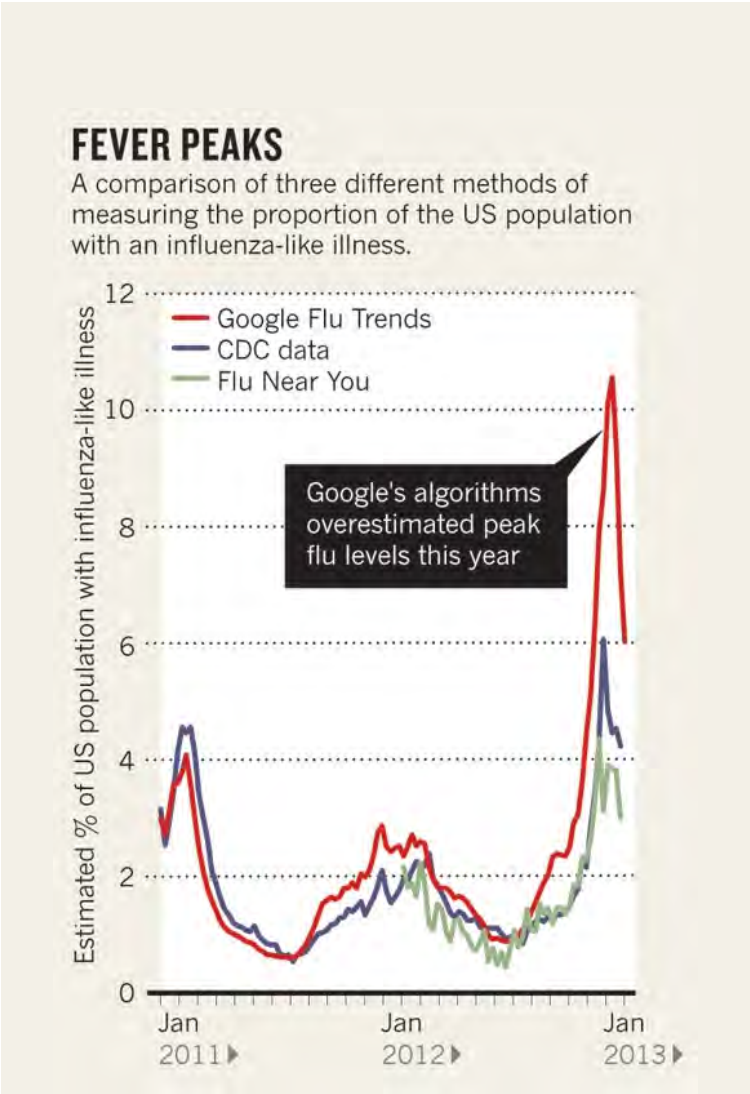


United States: Influenza-like illness (ILI) data provided publicly by the U.S. Centers for Disease Control.

Source: http://www.google.org/flutrends/

When Google got flu wrong

"The problems may be due to widespread media coverage of this year's severe US flu season [...] The press reports may have triggered many flu-related searches by people who were not ill."



Source: http://www.nature.com/news/when-google-got-flu-wrong-1.12413

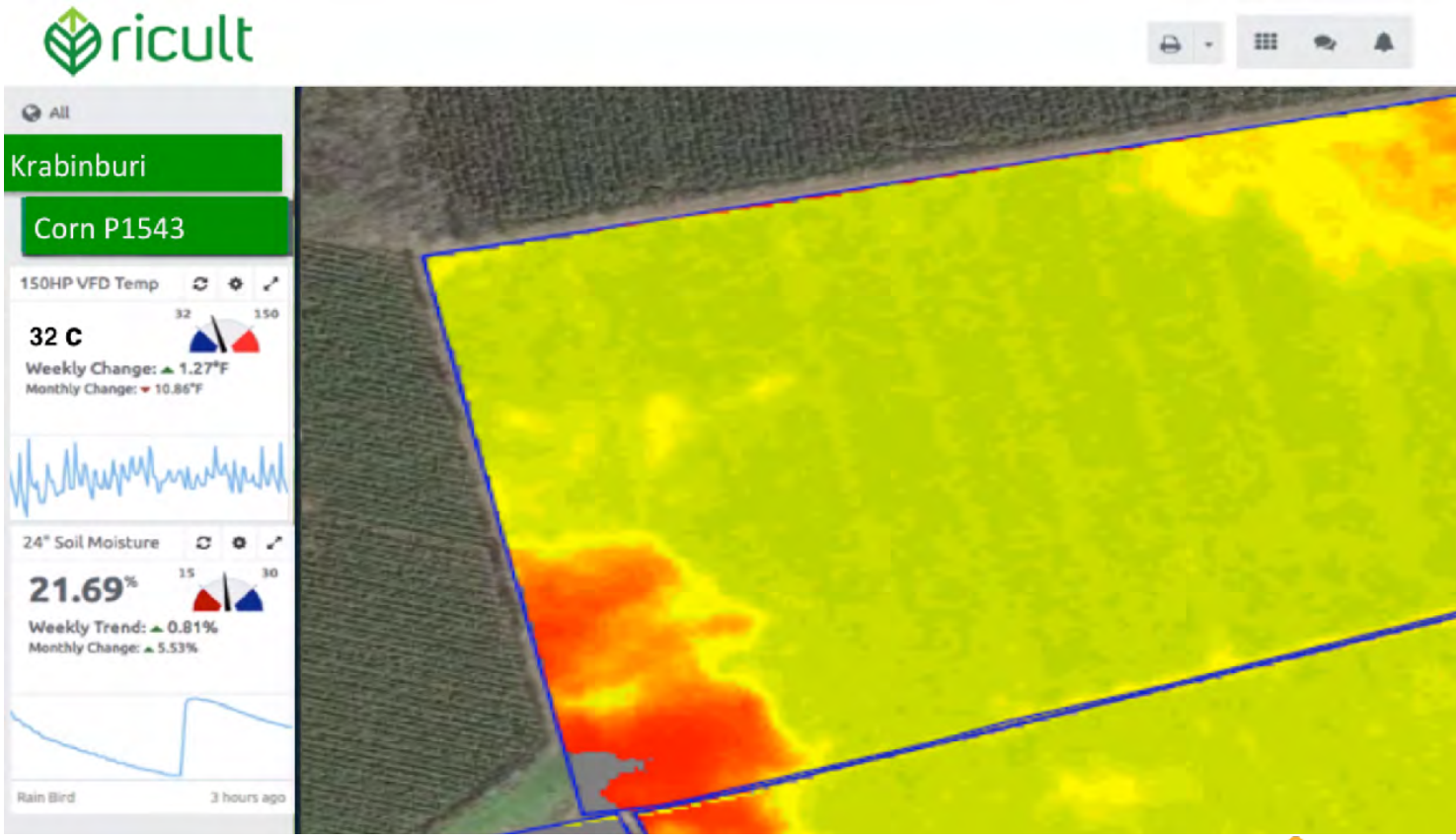
Facebook's Population Map

Unprecedentedly high-resolution population maps

Source: https://code.facebook.com/posts/1676452492623525/connecting-the-world-with-better-maps/

Ricult

Ricult brings the forefront system that provides farmers access to cheap, non-exploitative loans, without the hassle of lengthy processes or any collateral requirements



Source: http://www.ricult.com/

ECONOMICS

Combining satellite imagery and machine learning to predict poverty

Neal Jean,[1,2*] Marshall Burke,[3,4,5*†] Michael Xie,[1] W. Matthew Davis,[4] David B. Lobell,[3,4] Stefano Ermon[1]

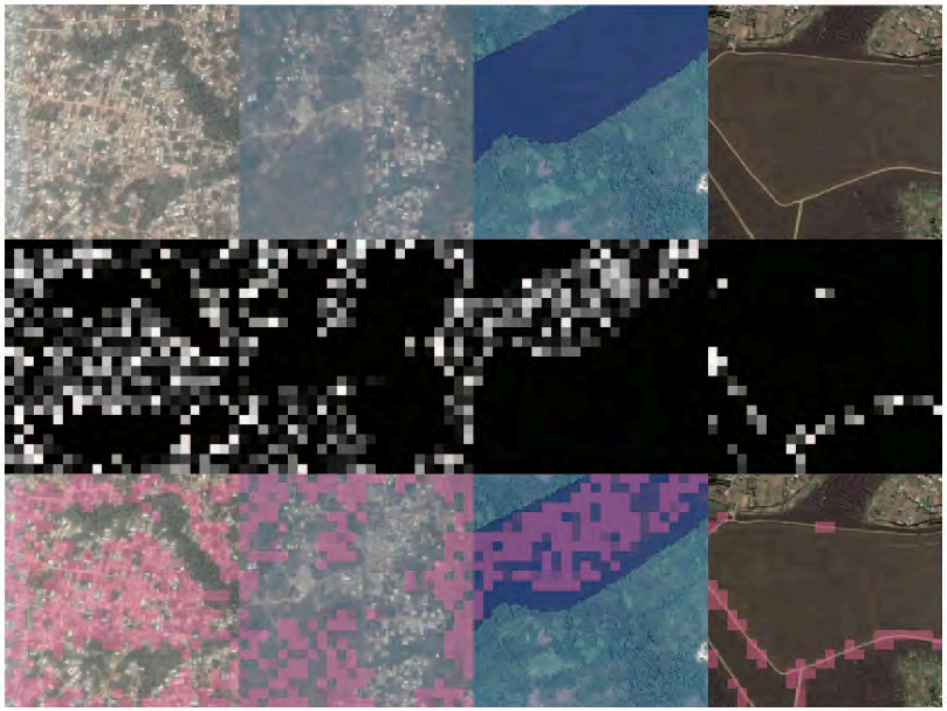


Fig. 2. Visualization of features. By column: Four different convolutional filters (which identify, from left to right, features corresponding to urban areas, nonurban areas, water, and roads) in the convolutional neural network model used for extracting features. Each filter "highlights" the parts of the image that activate it, shown in pink. By row: Original daytime satellite images from Google Static Maps, filter activation maps, and overlay of activation maps onto original images

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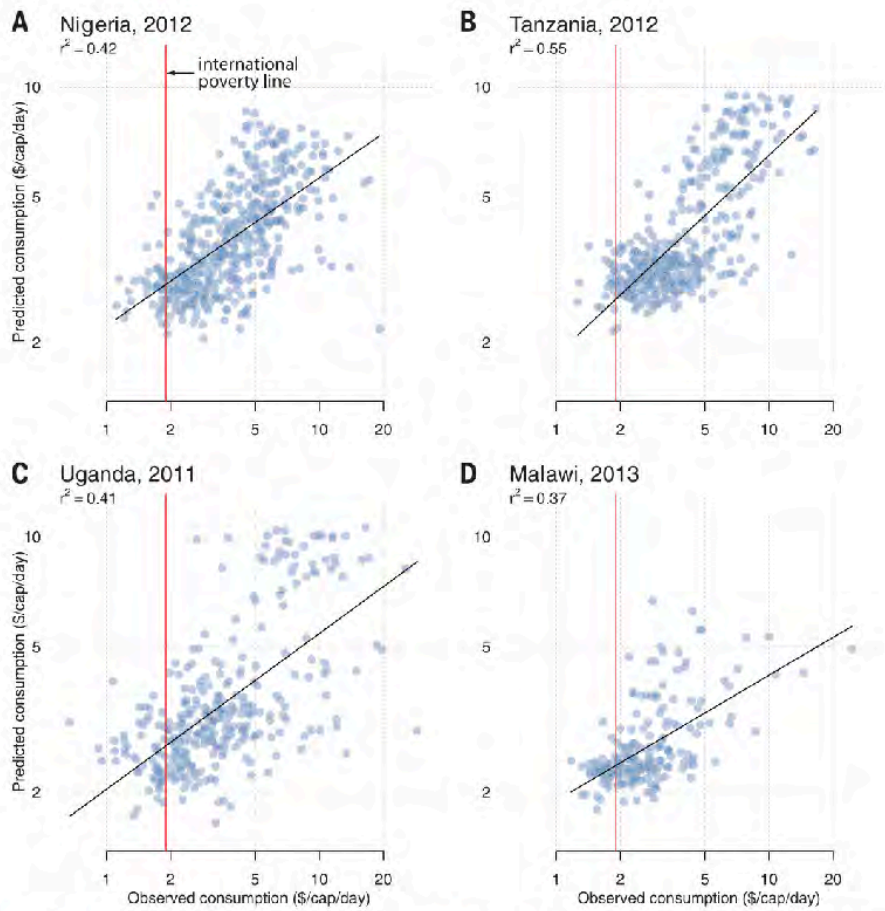

Fig. 3. Predicted cluster-level consumption from transfer learning approach (*y* axis) compared to survey-measured consumption (*x* axis). Results are shown for Nigeria (**A**), Tanzania (**B**), Uganda (**C**), and Malawi (**D**). Predictions and reported r^2 values in each panel are from fivefold cross-validation. Black line is the best fit line, and red line is international poverty line of $1.90 per person per day. Both axes are shown in logarithmic scale. Countries are ordered by population size.

ECONOMICS

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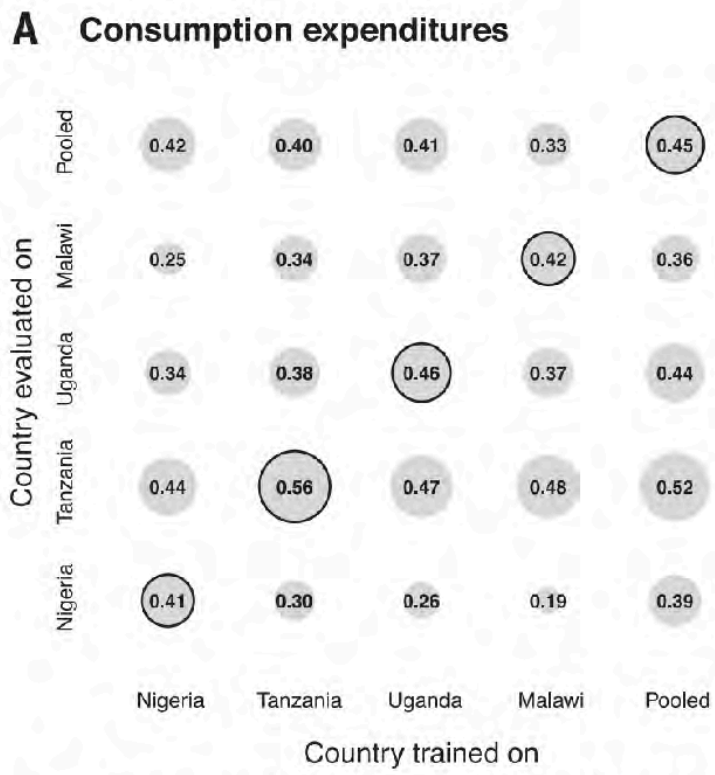
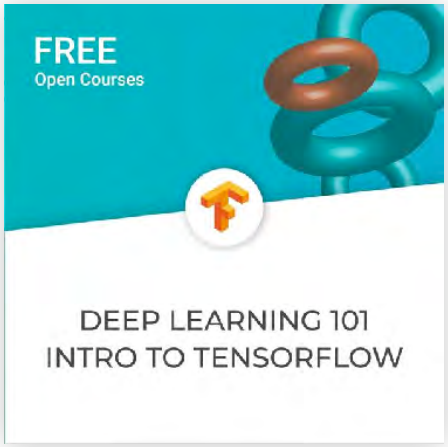


Fig. 5. Cross-border model generalization. (A)

Contract Intelligence

At JPMorgan Chase & Co., a learning machine is parsing financial deals that once kept legal teams busy for thousands of hours.

Source: https://futurism.com/an-ai-completed-360000-hours-of-finance-work-in-just-seconds/



www.skooldio.com

PIER

MAJOR ADVANCEMENTS IN MACHINE LEARNING